



AI ENHANCED ARRHYTHMIA DETECTION IN ECG DATA



A MINI PROJECT I REPORT

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ABSTRACT

In an age of intelligent healthcare and IoT-driven innovation, continuous monitoring of the vital signs and their analyzing importance is becoming a necessity. The present work is for developing and deploying a real-time heart rate sensing and categorizing system via an embedded machine learning mechanism. The proposed solution encapsulates within it a compact and efficient hardware-software architecture

eligible to run onboard or remotely in wearables health devices or home monitoring systems

The present system is based on the low power-consumption combined with Wi-Fi and Bluetooth connectivity and considerable processing capabilities of the ESP32 Wroom-32 microcontroller to run lightweight machine learning models. To this microcontroller is joined the MAX30100 pulse oximeter sensor to capture the PPG signals from the fingertip of the user before calculating the heart beats per minute (BPM), which acts as a critical indicator for assessing cardiovascular health.

The BPM values obtained from the sensor will be classified into three categories: Bradycardia (low heart), normal, and tachycardia (high heart). The classification is performed by a compact neural network model engine that has been trained and deployed using Edge Impulse, the ML platform. The data used for training the model has been collected from the MIT-BIH Arrhythmia Dataset. Rather than using the ECG signals as obtained, the pipeline trains on the RR interval that lies between any two successive R-peaks for estimating BPM values.

For this collection of data, RR interval detection algorithms were implemented in Google Colab, which then derived BPM values through ECG recordings, labeled according to clinical standards, and uploaded onto Edge Impulse. Edge Impulse carries out automated feature extraction and model optimization with onboard-friendly designs. The correctly trained model was exported as a C++ library and linked into the ESP32 development environment through Platform IO within Visual Studio Code.

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LIST OF ABBREVIATIONS

BPM	Beats Per Minute
ECG	Electrocardiogram
PPG	Photoplethysmography
ML	Machine Learning
AI	Artificial Intelligence
I ² C	Inter-Integrated Circuit
ADC	Analog-to-Digital Converter
OLED	Organic Light Emitting Diode
ESP32	Espressif Systems Microcontroller
CSV	Comma-Separated Values
MCU	Microcontroller Unit

CHAPTER 1

INTRODUCTION

However, in this era of smart healthcare, detecting cardiovascular-related abnormalities through real-time monitoring of vital signs is becoming increasingly important. We will develop an intelligent heart rate classification system for the ESP32 microcontroller, which implements MAX30100 (a Pulse Oximeter and Heart Rate Sensor), and a machine learning model built in Edge Impulse. The MAX30100 continuously reads the heart rate of the user in beats per minute (BPM), and this value is read by the ESP32 using I2C communication. These BPM values are then sent over to a pre-trained ML model hosted on Edge Impulse that divides the heart rate into 3 categories (Bradycardia, Normal, Tachycardia) The classification results can be shown on an OLED screen or sent to a mobile application or web dashboard to give immediate feedback. Such integration of IoT and AI serves as a compact, low-cost, and efficient solution for continuous health monitoring, which can be employed in wearable health devices, fitness trackers, or home-based medical systems.

1.1 PROJECT OVERVIEW

We propose to create a smart and portable heartrate measuring system which not only measures the users heart rate but also classifies the heart behavior using machine learning for better health experience. By incorporating a MAX30100 pulse oximeter sensor that detects heart rate, an ESP32 microcontroller to process the data, and a machine learning model based on Edge Impulse, this system has the capability of accurately detecting and classifying the heart rate into 3 categories: Bradycardia (low heart rate), Normal, and Tachycardia (high heart rate).heart rate, captured by the MAX30100 sensor in real-time, processed by the ESP32 and passed through the

trained model. The classification result can be displayed on an OLED or sent to a mobile app or cloud dashboard for remote health tracking and can be viewed over the internet. This project leverages the potential of embedded systems, biometer, and Deep Learning algorithm and presents a dependable and user- friendly continuous cardiac monitoring solution in the healthcare, fitness, and IoT- based wellness ecosystem. More Like This Short Time Novel Punches Up sci- fi Stories

As previously mentioned, one possible use of this classification method would be to alert those who have normal heart rates to see the doctor in a timely manner. This demonstrates what is achievable through edge computing combined with machine learning, which enables rapid tracking, and privacy since you aren't required to upload private health data to the cloud for processing. As the ESP32 has inbuilt Wi-Fi and Bluetooth support, this can scale further with real-time notification via mobile apps or smart watches in future. The modular architecture provides an opportunity for the system to be extended by simply adding extra sensors or can be adapted for measurement of other physiological parameters making it a scalable platform for greater health monitoring solutions. Across the spectrum, the project is emblematic of the increasing trend in AI-powered, IoT- enabled personalized care, serving as an approachable yet innovative leap toward appropriate and preventive health technology.

CHAPTER 2

LITERATURE SURVEY

The use of machine learning algorithms in heart health monitoring has led to significant improvements in the domain of biomedical signal processing. Electrocardiogram (ECG)-based classification algorithms have been more precise in detecting cardiac abnormalities like arrhythmia. Besides enhancing diagnosis accuracy, utilization of these models helps in the development of portable, affordable health monitoring systems. The technique and design of the proposed heart rate classification system using Edge Impulse and ESP32 have been influenced by extensive research contributions, which are discussed in this literature review.

2.1 INTRODUCTION

With the increase in the number of cardiovascular diseases (CVDs), the need for the early but precise diagnosis of heart conditions is most felt. Besides a host of cardiac issues, arrhythmia or irregular heartbeats are important indicators of perhaps fatal conditions. Though electrocardiograms (ECGs) are the most common methods by which cardiac activity is monitored, they require a long time for manual interpretation, hence introducing human errors. This study touches upon the automation of cardiac arrhythmia detection and classification on advanced machine learning (ML) and deep learning (DL) methodologies. In addition, large publicly available datasets like PhysioNet Challenge 2017 and MIT-BIH Arrhythmia Database are used to implement models with optimizations such as Bidirectional Long Short-Term Memory (Bi- LSTM) for binary classification and 1D Convolutional Neural Network (1D-CNN) for multiclass classification. Models could very well differentiate between normal and abnormal heart rhythms, where the complex familiar conditions such as atrial fibrillation and all forms of bundle branch block are included.

2.2 BACKGROUND

Pokharel, Atit, Dahal, Shashank, Chhetri, Prof. Dr. Bhupendra Bimal, and Sapkota, Pratik (2023) ‘Electrocardiogram (ECG) Based Cardiac Arrhythmia Detection and Classification using Machine Learning Algorithms’, International Conference on Technologies for Computer, Electrical, Electronics & Communication , Vol. 4, pp. 7352–7451.

It studies the application of ML and DL techniques for the detection and classification of cardiac arrhythmias based on electrocardiogram (ECG) signals. The authors developed a machine-learning model with the aim to classify accurately arrhythmic ECG signals. Data from PhysioNet and MIT-BIH databases has been utilized in this work for the processing of ECG signals. Initially, the work was carried out for binary classification: training a Bidirectional Long Short-Term Memory (Bi- LSTM) model to distinguish normal from atrial fibrillation signals. So, an important point in the study survey of doctors to get their views on the applicability and improvement areas of AI- based ECG classifiers. Meanwhile, the survey results were a clear indication for these systems and presented better ideas for improved future models. In furtherance of this feedback, the authors constructed a Convolutional Neural Network (CNN) system. 1 It was aimed at improving the accuracy of classifying five different types of ECG signals. 1 A rigorous model evaluation was carried out under stratified 5-fold cross-validation. 2 A web portal with the trained model for real- time classification demonstrates the practicality of applications in the real world. 3 Such models have potential uses for several healthcare applications, including remote health monitoring, predictive healthcare, assistive diagnostic tools, and educational training

Atshaya, P., Sankaran, V., and Varalakshmi, S. (2023) ‘An improved hybrid AI model for prediction of arrhythmia using ECG signals’, Biomedical Signal Processing and Control, Vol. 80, 104318.

It also provides a solid hybrid artificial intelligence (AI) technology that would, with great accuracy, predict and classify types of cardiac arrhythmias generated through electrocardiogram (ECG) signals. This very system resolves the core issue of differentiating normal heart rhythm versus that of an abnormal one, which is imperative in the early diagnosis and treatment of cardiovascular diseases. To do evaluation, authors utilized the MIT-BIH Arrhythmia Database, a benchmark data that consisted in 47 subjects of already annotated ECG traces. Preprocessing activities of denoising using Discrete Wavelet Transform (DWT) processes the ECG data followed by segmentation and resampling to ensure balanced classes for training. Bidirectional Long Short-Term Memory (BiLSTM) is chosen as deep feature extractor due to its capacity to capture temporal dependencies in time-series data from past and future perspectives. Principal Component Analysis (PCA) optimizes these features to reduce dimensionality, increase speed for processing, and ultimately improve model generalization. For classification purposes, the various evaluated machine learning models include Random Forest, SVM, Decision Trees, and KNN, where Random Forest is said to give the best results among all evaluated models. The score described by the combination of BiLSTM plus Random Forest plus PCA is 98.46 percent accuracy, about 98 percent high sensitivity and specificity, with a very low prediction time meanwhile (0.089seconds).

Singh, Amrit, Mahapatra, Harisankar, Biswal, Anil Kumar, Mahapatra, Madhumita, Singh, Debabrata, and Samantaray, Milan (2024) ‘Heart Disease Detection Using Machine Learning Models’, Procedia Computer Science, Vol. 235.

Cardiovascular diseases are responsible for high mortality and morbidity, thus making heart disease indeed a major global issue. Early recognition and correct diagnosis are key in maximizing patient outcome and minimizing mortality. Machine learning (ML) techniques have gained considerable attention as adjuncts for healthcare professionals in the diagnosis of cardiac ailments that can be both timely and accurate. This paper series study covers the supervised machine-learning methods, such as Decision Tree Theory (DT), Random Foresting (RF), Support Vector Machine (SVM), and Principal Component Analysis (PCA) for heart disease detection. The methodology is characterized by data collection and entries (data pre- processing, data processing, feature selection, model training, and performance evaluation), and among these, data pre-processing stands out. In this paper, heart disease is predicted for 10 years using a dataset called Framingham. The process of data pre-processing involves the handling of missing values and expanding the dataset size using random oversampling techniques. The performance of prepared ML models is measured using different metrics like accuracy, precision, recall, F1 score, and AUC ROC. Results indicate that the Random Forest gives the maximum accuracy (97%) compared to Decision Tree (84%), PCA (79%), and SVM (68%). It can be concluded from this research that machine-learning techniques, especially Random Forest, can effectively aid in the detection of heart disease, thereby improving diagnostic accuracy.

Kumar, S., Bhushan, B., Bhambhu, L., Singh, D., and Choubey, D.K. (2023) ‘A Deep Learning-Based Model for Arrhythmia Detection Using Feature Selection and Machine Learning Methods’, Springer Nature Singapore Pte Ltd, Vol. 5, pp. 7345-7451.

The paper reviews the use of deep learning, with special emphasis on CNNs, in the detection of arrhythmia, an irregularity in heartbeats formation and the largest amorphous killer worldwide. The authors have investigated the feasibility of pre-trained CNN models such as AlexNet, VGG-16, and VGG-19 for feature extraction using the MIT arrhythmia dataset, followed by classification using different machine learning algorithms such as decision trees, k-nearest neighbors (KNN), linear discriminant analysis (LDA), and support vector machines (SVM). The study further improved model accuracy by feature selection using minimum redundancy maximum relevance (mRMR). From their experimental results, it shows that the CNN model as proposed along with the mRMR feature selection algorithm exhibits a classification accuracy of 99.46%, demonstrating the power of deep learning in annulling arrhythmia. The second paper, "Heart Disease Detection Using Machine Learning Models", emphasizes the role of machine learning models in detecting heart disease, which is a major public health challenge across the globe. The study applies the Framingham dataset and evaluates the performance of supervised machine learning methods using Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), and Principal component analysis (PCA). The research addresses the overfitting problem on heart disease detection models by applying several machine learning techniques with preprocessing steps such as data cleansing, normalization, and feature selection. Results indicate that the Random Forest algorithm gives 97% accuracy, thus proving the worthiness of machine learning in heart disease detection.

Both papers focus on the importance of machine learning in medicine, especially in heart-related disease detection. The first paper, "Deep Learning-Based Model for Arrhythmia Detection...", utilizes deep learning techniques and feature selection methods for the accurate detection of arrhythmia and achieves high classification accuracy. The second paper, "Heart Disease Detection Using Machine Learning Models," investigates machine learning algorithms in heart disease detection, where Random Forest seems to yield the most optimistic result. Both studies illustrate the promise of machine learning in improving the accuracy and efficiency of diagnosis in critical cardiovascular conditions.

Khandoker, A.H., Mohammad, B., Saleh, H., Ismail, M., Bayasi, N., and Tekeste, T. (2016) 'Low-Power ECG-Based Processor for Predicting Ventricular Arrhythmia', IEEE Access, Vol. 4, pp. 7462–7471.

It describes an architecture for completely integrated electrocardiogram (ECG) signal signal processing for ventricular arrhythmia prediction based on some values of ECG features with an initial stage of implementation. Techniques thus focus on "detecting" and "delineating" the P-QRS-T waves and thus extracting the fiducials within real-time adaptive mode. These techniques should ever be robust in changing the ECG signal, and should have a highly sensitive and precise detection of key features. Validation of testing the effectiveness of this processor has been made possible through the heart signal recordings between these two databases controlled by MIT PhysioNet and American Heart Association. Application specific integrated circuit simulation results have also proven that this processor achieved 86% classification accuracy on out-of-sample validation data using a window size of 3 seconds. Ventricular arrhythmia causes about 300,000 deaths a year because of abnormal rhythmic ECG pattern leads to ventricular arrhythmias and are responsible for around 75%- 85 % sudden deaths in heart patients without instant treatment. The most common causes of these arrhythmias include coronary heart disease, hypertension or cardiomyopathy, and

they elucidate clear clinical diagnoses and timely management.. The paper introduces two important types of ventricular arrhythmias namely ventricular tachycardia (VT) and ventricular fibrillation (VF). It is a fast rhythm of three consecutive beats arising from the ventricles and having a rate in excess of 100 beats/min that defines VT, whereas chaotic activation of the ventricles is termed VF and results in immediate cessation of blood circulation. Implantable cardioverter defibrillators may protect people from dying suddenly because of ventricular arrhythmia; however, most sudden deaths occur in the "low risk" category. For diagnosis of ventricular arrhythmias, long-term ECG must be interpreted. This requires analyzing 12-lead ECGs for changing characteristics of the signal for an assessment of possible ventricular arrhythmias. The actual detection of ventricular arrhythmia is based on attributes from the P- QRS-T components' intervals, amplitudes, and morphology. Many feature- extraction methods, morphological, spectral or mathematical professional methods, and machine learning methods, such as neural networks and support vector machine (SVM) models, have been developed for detecting ventricular arrhythmia from ECG signals. However, these techniques mostly have long intervals of detection, resulting in delays in the detection of the actual condition since late detection intervals may not allow timely intervention.

Kumar, M., Rai, A., Surbhit, and Kumar, N. (2023) ‘Autonomic edge cloud assisted framework for heart disease prediction using RF-LRG algorithm’, Multimedia Tools and Applications, Vol. 83, pp. 5929–5953.

The design of a fully integrated electrocardiogram (ECG) signal processor (ESP) for ventricular arrhythmia prediction based on a specific set of ECG features and a naïve Bayesian classifier is presented in this paper. The investigation focused on real-time adaptive methods for detection and delineation of the P-QRS-T which or how they extract fiducials.

available between the MIT PhysioNet and the American Heart Association databases. Application specific integrated circuit (ASIC) simulation results showed the performance of the processor as achieving 86% accuracy in classification on out-of-sample validating data, obtained via window size. Sudden cardiac death is now the most essential research in relation to ventricular arrhythmia and occupies approximately one-third of the mortality in any one year in the U.S. An irregular rhythm of the ECG contains ventricular arrhythmias and causes around 75% to 85% for sudden death to heart patients without immediate treatment. Some of the more common sources of arrhythmias are coronary heart disease, hypertension, or cardiomyopathy. The paper mentions two significant varieties of ventricular arrhythmias: ventricular tachycardia and ventricular fibrillation. It defines VT, which is fast rhythm arising from the ventricles, defined by three consecutive beats having a rate of more than 100 beats/min; while VF is defined as chaotic activation of the ventricles away from physiologic pattern occurring with immediate cessation of blood circulation. They may access most sudden deaths because of low risk; however, implantable cardioverter-defibrillators may save many people from dying suddenly from ventricular arrhythmias. Diagnosis requires long-term ECG to interpret the readings. Such patients have to have long-term ECG interpreted and analyzed for changes in signal characteristics to assess possible pathology involving ventricular arrhythmias using a 12-lead ECG. There are features associated with normal ventricular arrhythmia from intervals, amplitudes, morphologies of P-QRS-T component signals. Many methods, morphological, spectral, mathematical professional methods, and machine learning methods have been developed for detecting arrhythmia of the ventricle from ECG signals, such as neural networks and support vector machine (SVM) models. However, these features are mostly characterized by long detection intervals, meaning that they delay detection of the actual condition as possible time intervention may not be allowed.

Application specific integrated circuit (ASIC) simulation results showed the performance of the processor as achieving 86% accuracy in classification on out-of-sample validating data, obtained via window size. Sudden cardiac death is now the most essential research in relation to ventricular arrhythmia and occupies approximately one-third of the mortality in any one year in the U.S. An irregular rhythm of the ECG contains ventricular arrhythmias and causes around 75% to 85% for sudden death to heart patients without immediate treatment. Some of the more common sources of arrhythmias are coronary heart disease, hypertension, or cardiomyopathy. The paper mentions two significant varieties of ventricular arrhythmias: ventricular tachycardia and ventricular fibrillation. It defines VT, which is fast rhythm arising from the ventricles, defined by three consecutive beats having a rate of more than 100 beats/min; while VF is defined as chaotic activation of the ventricles away from physiologic pattern occurring with immediate cessation of blood circulation. They may access most sudden deaths because of low risk; however, implantable cardioverter-defibrillators may save many people from dying suddenly from ventricular arrhythmias. Diagnosis requires long-term ECG to interpret the readings. Such patients have to have long-term ECG interpreted and analyzed for changes in signal characteristics to assess possible pathology involving ventricular arrhythmias using a 12-lead ECG. There are features associated with normal ventricular arrhythmia from intervals, amplitudes, morphologies of P-QRS-T component signals. Many methods, morphological, spectral, mathematical professional methods, and machine learning methods have been developed for detecting arrhythmia of the ventricle from ECG signals, such as neural networks and support vector machine (SVM) models. However, these features are mostly characterized by long detection intervals, meaning that they delay detection of the actual condition as possible time intervention may not be allowed.

Kumar, A., and Rao, B.M. (2022) ‘Detection of heart arrhythmia based on UCMFB and deep learning technique’, *Sadhana*, Vol. 47, Article 252.

The task of discerning cardiovascular diseases (CVDs) through the interpretation of ECG signals is among the most challenging on this earth. The authors advance a novel concept-an automated hybrid system. They emphasized that with the rising mortality due to heart-related diseases, there is a need for early detection and timely diagnosis in emergencies. They argued that the capability of DL techniques could revolutionize the detection of cardiovascular disorders, thereby aiding cardiologists in their difficult manual examination of ECGs. Their proposed system is based on the application of UCMFB to ECG signal decomposition for identification of irregular heartbeats, alongside performing Wavelet Transform Packet (WTP) for the conversion of these signals into 2D images. These 2D images would be undergone by the deep learning model ResNet-50, whose output is customized to classify a sample into four categories: healthy, Atrial Fibrillation (AF), Congestive Heart Failure (CHF), and Hypertension (HT). The patient data for any cardiovascular condition considered in this work come from the NSR database and BIDMC CHF database. These will then be UCMFB-preprocessed to remove noise and next will decompose the signal into separate sub bands such that useful results are attained for feature extraction and further analysis. Segments extracted from preprocessed signals are first put together in short (2 seconds and 5 seconds) and long (5 minutes and 8 minutes) time frames of data, which are then WTP-transformed into 2D images. ResNet-50 identifies different cardiovascular conditions using its deep learning feature through extracting and classifying such images using a convolutional neural network (CNN). DL classifier, significantly enhancing model performance.

Hernandez, C., Mitchell, D.M., Rosario, N., and Lipsitz, S.R. (2023) ‘Single- lead arrhythmia detection through machine learning: cross-sectional evaluation of a novel algorithm using real-world data’, *Journal of Electro cardiology*, Vol. 75, pp. 45–52.

A hybrid machine learning model has been used in this study to test the classification of eight common arrhythmias with single-lead ECG data, particularly from acutely ill patients in a home hospital setting. Such an approach is not rigorous with respect to the real-world consequences, and, therefore, its findings are of importance. This is important because of the need for accurate and timely detection of arrhythmia for patient care. The model was trained on a large single-lead ECG database and then carried out a test on data collected from patients admitted for treatment at two Boston hospitals from June 2017 to November 2019. This hybrid approach, combining traditional signal processing and machine learning, was aimed at achieving accurate arrhythmia detection from real-world ECG data, which may be noisy because of physiological interferers. The study demonstrated that various arrhythmias would be classified efficiently using the hybrid machine learning model. The model analyzed more than 2.6 million minutes of single-lead ECG data from 423 patients and detected 691,478 arrhythmias. The overall accuracy, sensitivity, and specificity were 98%, 98%, and 100%, respectively, for the detection of any arrhythmia. Among the arrhythmias, excellent performance was detected for most; however, pauses were diagnosed with the most accuracy, while paroxysmal supraventricular tachycardia was diagnosed with slightly less. The model's false-positive rate for any arrhythmia was very low at 0.2%, but its false-negative rate was 1.9%. Altogether, this study suggests hybrid machine-learning models are promising for categorizing cardiac arrhythmias from single-lead ECGs in real- world conditions. The model's high accuracy and low false-positive rate make it a possible candidate for use in improving arrhythmia detection in acutely and chronically ill patients.

El-Sofany, H.F. (2024) ‘Predicting Heart Diseases Using Machine Learning and Different Data Classification Techniques’, IEEE Access, Vol. 12, pp. 123456–123467.

This study investigates machine learning (ML) and empirical approaches to predicting heart disease by using a combined suite of private and publicly available datasets. The aim is to develop an accurate ML methodology by adopting different feature selection methods such as chi-square and analysis of variance (ANOVA) and mutual information (MI) to find the features most relevant for prediction purposes. Ten different ML classifiers (Naive Bayes, SVM, XGBoost, etc.) are used to evaluate the efficacy of the selected features and settle on an optimal classification technique. The crux of the research is towards the important question of detecting heart disease at an early stage, thereby significantly reducing the mortality rates. Using ML algorithms, the authors hope for a more precise and data-driven diagnosis that can perhaps overcome the limitations of traditional methods, which may not be accurate or viable for continuous monitoring of patients. Their ultimate aim is to create a tool for healthcare workers and common citizens to assess the risk of heart diseases in an affordable and timely way. The research work further gives birth to a mobile app that will allow users to get an instant heart disease prediction based on signs and symptoms they enter. The app will generate instant heart disease prediction based on signs and symptoms entered by the user. The ML models that proved to give the best performance during the investigation will, therefore, be integrated into this app, showing how ML could be channeled towards real-world applications which may bring about the improvement of healthcare outcomes. Explainable Artificial Intelligence is further highlighted using the SHAP methodology to enhance the transparency and trustworthiness of the predictive model; thus, healthcare providers will be equipped to comprehend and validate the recommendations from the system.

Petrillo, A., Liotto, C., Toscano, G., Santini, S., and Tufano, V. (2022) ‘A multiclass CNN cascade model for the clinical detection support of cardiac arrhythmia based on subject-exclusive ECG dataset’, Computers in Biology and Medicine, Vol. 145, 105390.

The proposed classification algorithm set out herein would enhance the detection accuracies of heart disorders using ECG waveforms. In this classification algorithm, two convolutional neural networks (CNN): a Binary CNN to detect arrhythmic heartbeats and a Multiclass CNN to identify the specific type of disorder are used in a cascade way. This cascade architecture rules-added data-splitting scheme enables the prediction of patients' health status for those not seen before. The training method is subject exclusive, which means ECG tracks are assigned into training, validation, and test sets based on patient ID and the distribution of heartbeat-disorder classes. Besides, pre-processing includes filtering noise from ECG data and normalizing it to secure the accuracy of its following classification. 1D- CNN layer building blocks were used to realize the CNN models; the Binary and Multiclass CNNs used different configurations for optimization in performance. Evaluation of the performance of the proposed algorithm was performed on the MIT-BIH Arrhythmia database, for which results of a classification accuracy of 96.2% were shown, indicating advantages of the cascade solution over a simple Multiclass CNN. The ability of the algorithm to detect and classify different heart rhythm disorders reliably embodies the potential for improvement in diagnosing and preventing cardiac pathologies, highlighted by its ability to detect PVC, LBBB, and RBBB.

2.3 SUMMARY

The reviewed literature emphasizes the development of various machine learning (ML), deep learning (DL), and hybrid computation models for the detection, classification, and prediction of cardiac arrhythmias and heart diseases. The key diagnostic input following electrocardiogram (ECG) signals in the studies is the detection of arrhythmias considered dangerous for the life of the cardiovascular systems. Multiple models, including BiLSTM, CNNs, ResNet, hybrid CNN-LSTM, and naive Bayes classifiers, have been deployed, most achieving exceptional accuracy rates of over 95%. Most hybrid models use traditional signal processing methods in conjunction with any ML method such as BiLSTM with Random Forest, or CNN cascades, and thus take the top scores in classification accuracy and robustness. Some take operational emphasis toward real-time or low-latency use cases, especially in edge-cloud integrated frameworks, utilizing timely diagnostics in smart healthcare systems. In addition, mobile and web-based applications to improve accessibility and real-life usability have also been in development. Some data feature selection techniques, PCA and wavelet transforms, SHAP methods for explainable AI, are highlighted for their contribution to the transparency and efficiency of the models. Models were distributed and validated against benchmark datasets (e.g., MIT-BIH, PhysioNet) and patient data collected in the real world, while further validation was ensured through an intensive cross-validation procedure to confirm their generalizability. Overall, these combined works lead to scaled, efficient, and accurate cardiac monitoring solutions to support predictive healthcare, remote patient monitoring, and quick diagnostics.

CHAPTER 3

METHODOLOGY AND WORKING PRINCIPLE

This chapter introduces the systematic process and technical background of the AI-Enhanced Arrhythmia Detection System. Biomedical signal processing, embedded systems, and machine learning are applied in the project to identify and classify the heart rate condition in real time. The methodology is such that the data is categorized into three main categories—bradycardia, normal, and tachycardia and accurate BPM (beats per minute) detection from ECG signals.

3.1 PROBLEM IDENTIFICATION

Heart disease is among the leading causes of death worldwide, and prevention and early treatment are based on the detection of abnormal heart rate conditions. Continuous and real-time heart rate monitoring is not readily available, especially in rural or underdeveloped communities. Traditional monitoring systems are bulky, expensive, or require hospitalization. Monitoring one's heart rate on a daily basis is therefore not possible. The world today requires a low-cost, portable, and intelligent system that can continuously monitor heart rate in real time and detect abnormal patterns such as bradycardia and tachycardia without constant monitoring by physicians.

3.2 OBJECTIVE

The aim of this project is to create and implement a real heart rate classifier system using an ESP32 microcontroller and a MAX30100 pulse sensor. The system will capture BPM data from the user and categorize the heart rate as bradycardia, normal, or tachycardia using a machine learning model on Edge Impuls

The intelligent solution is designed to provide accurate, affordable, and easily accessible heart monitoring that can assist individuals and healthcare professionals in detecting and addressing potential heart-related complications at an early stage.

3.3PRINCIPLEOF THE SYSTEM

The heart rate monitoring system is founded on the fundamental principle of detecting the heartbeat or pulse optically or electrically and digitizing the bio-signal to process, analyze, and classify. In implementing this project, we utilize a MAX30100 sensor, which integrates two fundamental components—a pulse oximeter and a heart rate sensor—to detect the changes in blood volume in the microvascular bed of tissue. The process of working begins with the photoplethysmography (PPG) technique. The MAX30100 sensor emits light through LEDs (red and infrared) onto the skin. Because the heart is pumping blood, the volume of blood within the capillaries varies, and with it, the volume of absorbed and reflected light. The reflected light is picked up by a photo detector in the sensor, and it is used to calculate the BPM. Since the blood flow fluctuates with each heartbeat, this fluctuation is utilized to calculate the beats per minute (BPM). Once raw sensor data has been collected, it is funneled to a microcontroller, in this case, the ESP32 WROOM 32. The sensor is a powerful chip can read analog values, process them, and determine the BPM in real time. The ESP32 also handles other tasks such as printing the heart rate to an OLED display. The system is enhanced with AI for detecting abnormalities. With the

aid of a pre-trained machine learning model, designed using Edge Impulse, BPM values are divided into three wider classes — bradycardia (low heart rate), normal, and tachycardia (high heart rate). Training data is taken from the MIT- BIH Arrhythmia Database, pre-processed using Google Collab for transforming ECG signals to BPM values, and utilized for model training. The model is exported as an Arduino library and uploaded to the ESP32 using VS Code and Platform IO. This allows the system to not only read and display but also semantically understand heart rate and sound an alarm when suspicious patterns are found.

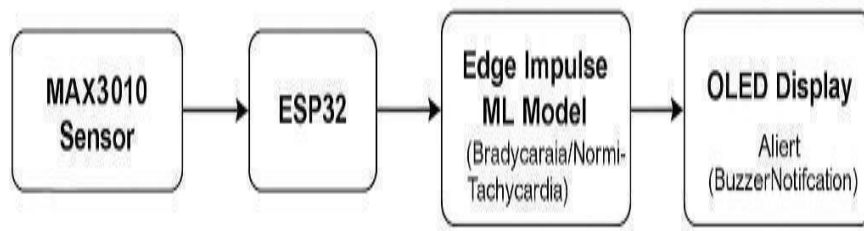


Figure 3.1 Methodology

3.4 WORKFLOW OF THE PROPOSED SYSTEM

The proposed system in this paper contains a systematic and multilevel process that starts from data collection and ends at real-time classification and display of heart rate information. The process is specifically designed to track the heart condition of the user correctly and alert the user in case of any irregularity. The system is organized into five core phases: collection of sensor data, data preprocessing, machine learning inference, results interpretation, and user feedback. The first component is capturing live heart rate data using the MAX30100 sensor. The sensor can capture pulse rate and blood oxygen, but in

this project, the pulse rate data is used. The user places his fingertip on the sensor where it emits red and infrared light that changes in blood volume with each beat of the heartbeat. The reflected light is translated into an electrical signal that produces the user's pulse waveform. After the generation of sensor data, it is fed into the ESP32 microcontroller.

The microcontroller is fed the analog data from the sensor and performs simple signal filtering and peak detection to determine beats per minute. The ESP32 was selected because it contains a dual-core processor and Wi-Fi support, thus making it the ideal candidate for real-time IoT and edge AI applications. The second is machine learning inference. A bespoke model is trained on the Edge Impulse platform. The model is trained to classify the heart rate into three categories, namely bradycardia, normal, and tachycardia. For this model's training, ECG data was extracted from the MIT-BIH arrhythmia database and processed using Python scripts from Google Collab, where the BPM values were read from the dataset and labelled based on medically accepted low, normal, and high heart rate thresholds. After training and testing the model, it is exported as an Arduino library. It is then imported into the ESP32 code via Platform IO inside VS Code. The ESP32-WROOM-32 performs the model inference locally, so that it can make the prediction of the heart condition locally on the device without needing an internet connection. Once the prediction has been made, the outcome is read and is shown on the connected OLED display. The display indicates the real-time BPM and the class to which it belongs. If the BPM value falls under bradycardia or tachycardia, the system can be made to initiate an alert in the form of a buzzer or send an alert over Wi-Fi to a smartphone app or web interface. In short, the system proposed here integrates embedded hardware and machine learning in a low power, smooth, and efficient way.

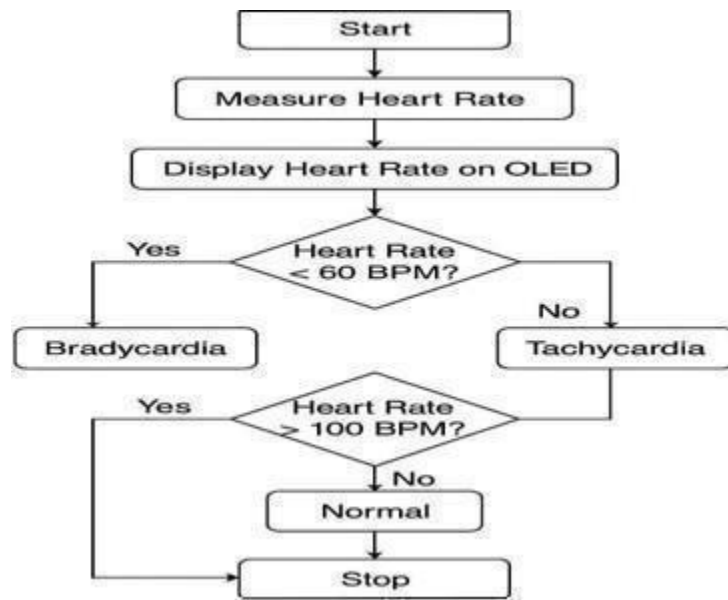


Figure 3.2 Workflow of the proposed system

3.5 HARDWARE DESCRIPTION

The hardware in this case is a collection of small, efficient, and robust pieces of hardware with defined functions in signal acquisition, processing, and display. It has at its core a microcontroller module that manages data acquisition and communication. It is equipped with in-built connectivity capabilities that enable easy data transfer and ease of use with software environments. In data acquisition and embedded sensor module that can sense very small changes in light is used, with the capability to provide accurate measurements of BPM values. An embedded display system is also provided to provide output in real- time from the hardware itself without the need for any external devices. All the devices are connected through a breadboard interface, which makes it easy to wire and debug during development of the machine learning model. Power is supplied directly through a USB line from a laptop, without the need for a power supply.

3.5.1 Microcontroller (ESP32-WROOM-32)

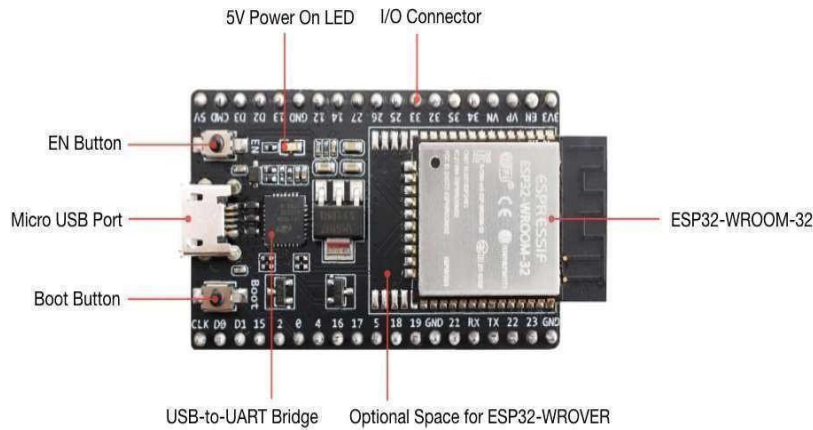


Figure3.3 microcontroller (ESP32-WROOM-32)

From the figure3.3, The ESP32-WROOM-32 is a high-performance microcontroller that is widely used in Internet of Things (IoT) use. It has a dual-core CPU, which allows it to execute tasks in parallel, thus improving its capability to execute multiple functions such as reading data from sensors and communication. The ESP32 has a built in Wi-Fi and Bluetooth module within it, which allows it to provide wireless connection to software and websites. The microcontroller also has a high number of GPIO pins that can be used to interface with external devices like sensors and displays. It also has a moderate level of memory, both RAM and flash storage, which makes it capable of executing complex jobs, including embedded machine learning models. It is also power-efficient, with multiple low-power modes supported by battery-powered devices. In this project, the ESP32 WROOM 32 is the master controller, which reads data from the MAX30100 sensor, processes it, and sends it to other systems for analysis or display.

3.5.2 Pulse and oxygen sensor (MAX30100)



Figure 3.4 MAX30100 pulse sensor

From the figure 3.4, MAX30100 is a pulse and blood oxygen level sensor and heart rate sensor that is intended to detect heart rate (pulse). The sensor uses photoplethysmography (PPG), a technique that detects changes in light absorption as blood flows through the vessels. It consists of two LEDs, namely infrared LED, red LED, and photodetector. The infrared and red light from the LEDs travel through the skin and the photodetector detects the reflected infrared light. These values allow the sensor to calculate the heart rate and SpO₂ level. MAX30100 is compact and powerful and efficient sensor, which makes it ideal for wearable health monitoring like the application used in this project, where it continuously monitors the user's oxygen level and heart rate.

3.5.3 OLED display



Figure 3.5 OLED display

From the figure 3.5, The OLED display module is a low-power, compact screen that uses organic light-emitting diode (OLED) technology to produce bright, crisp images with exceptional quality. Unlike traditional LCDs, OLED panels do not need a backlight, as each pixel generates its own light. This allows for true blacks, higher contrast ratios, and better energy efficiency. In this project, the OLED display is employed to show real-time health data collected from the MAX30100 sensor, including heart rate and blood oxygen saturation (SpO2) levels. Due to its small form factor, the OLED is easily integrated with the ESP32 microcontroller, making it ideal for portable and wearable health-monitoring applications. The instant visual feedback provided by the display enhances the user experience, allowing individuals to monitor their vital signs in real time. Additionally, the low power consumption of the OLED module helps extend battery life in mobile applications, contributing to the overall efficiency and convenience of the system.

3.6 SOFTWARE DESCRIPTION

The AI Improved Arrhythmia Detection System software comprises several phases, i.e., data preprocessing, machine learning model training, and deployment. Data preprocessing is carried out using Google Colab, where raw ECG signals of the MIT-BIH dataset are pre-processed to transform them into beats per minute (BPM) by detecting heartbeats and calculating intervals. The data is then uploaded to Edge Impulse, a machine learning model development platform. The features related to the BPM data are extracted neural networks algorithm are used to train the model. The trained model is then verified for accuracy. Then the model is deployed into the microcontroller using VS Code and Platform IO, where the machine learning model is deployed into the ESP32 microcontroller. which provides real-time heart condition classification from sensor data with immediate feedback through an OLED display.

3.6.1 Training machine learning model in edge impulse

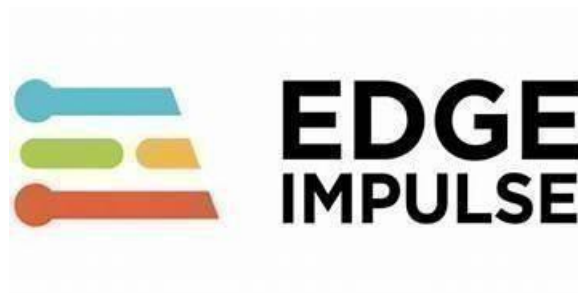


Figure 3.6 Edge Impulse

From the figure 3.6, Edge Impulse is used for training a machine learning model that classifies heart rate data into normal, bradycardia, and tachycardia categories. The first step involves uploading data into data acquisition, where in the preprocessed BPM data of Google Colab is uploaded. After that the platform automatically

calculates features, statistical such as mean, variance, and frequency components from the BPM data, which can be used to differentiate among various heart rate conditions. At the model selection and training step, an appropriate learning algorithm such as a decision tree or neural network is selected and trained using the labeled feature set. This is then tested and evaluated, wherein the performance of the model is confirmed using independent test data to verify accuracy and reliability. The confirmed trained model is then exported in the form of an Arduino library to render it deployable over the ESP32 microcontroller in embedded systems.

3.6.2 Model deployment in VScode (platform io with esp32)

The deployment stage includes embedding the trained machine learning model in the embedded environment through Visual Studio Code with the Platform IO extension. The Edge Impulse-exported Arduino library is imported into the Platform IO project. The ESP32 is programmed to retrieve live BPM data from the MAX30100 sensor and send it through the trained model in real-time. Based on the output class, the system determines whether the heart condition is normal, bradycardia, or tachycardia. Platform IO provides a robust environment for managing libraries, writing firmware, and uploading the code to the ESP32 board. This integration allows the machine learning model to function directly on the microcontroller, enabling real-time classification and display of heart rate status on the OLED module without requiring cloud-based processing.

3.7 IMPACT OF THE PRODUCT

This plays an important role in health monitoring because of its real-time, precise identification of heart conditions. It is a convenient, portable solution for people to keep track of their heart health, particularly for patients with heart conditions such as bradycardia, tachycardia, or normal heart rhythm. Using a low-cost, small device, the system makes heart rate monitoring available to the masses, which can potentially avert heart-related emergencies through early detection. The application of machine learning and embedded systems also makes the device able to precisely classify heart conditions without the need for continuous cloud connectivity. This makes the device perfect for daily use, providing peace of mind and enhancing overall health awareness.

3.8 UNIQUENESS AND FEATURES

This product stands out due to its utilization of embedded systems and real-time machine learning in classifying heart rates. In contrast to common heart monitors that provide mere data, this solution classifies heart conditions like bradycardia, normal, and tachycardia in an intelligent way, providing more insight into an individual's health. Utilization of MAX30100 sensor for pulse and oxygen level measurement with an ESP32 microcontroller for in-device processing that makes the solution more compact and power-effective.

3.9 SUMMARY

AI-Powered Arrhythmia Detection System is an intelligent health monitoring system that employs embedded systems and machine learning to classify heart rate conditions in real-time. The system employs the ESP32 WROOM 32 microcontroller to capture BPM data from the MAX30100 pulse and oxygen sensor. Local data processing is done using a machine learning model, which has been trained on Edge Impulse. This enables the system to accurately detect and classify bradycardia, normal, and tachycardia conditions without the use of cloud services. The real-time feedback in the OLED display ensures ease of monitoring of health. Google Colab carries out data preprocessing using ECG to BPM conversion to enable clean and reliable input for training. The system is cost-effective, portable, and internet-connection-independent, making it suitable for home or wearable health applications. The system provides real-time notifications and encourages early detection of heart conditions, making it effective and easy as a solution for personal health monitoring.

CHAPTER 4

DESIGN AND MODELLING

4.1 DESIGN:

AI based Arrhythmia Detection System is an intelligent, portable health monitor that detects and diagnoses irregular cardiac conditions in real time. It is designed on the ESP32-WROOM-32 microcontroller and makes use of the MAX30100 pulse and oxygen sensor to track vital health metrics such as heart rate (BPM). Processing is done locally in the device by an Edge Impulse-trained machine learning model without the need for ongoing internet connectivity or off-device processing. The system is supplemented by an OLED display to give real-time feedback to the user, graphically indicating heart rate status as bradycardia, normal, or tachycardia. Preprocessing is achieved by Google Colab to make the input to training clean and trustworthy. The entire setup is small, power-efficient, and appropriate for wearable or portable use. Easy to use and accessible, the system facilitates early cardiac abnormality detection and supports personal health monitoring. Its smart design makes it a great option for contemporary healthcare solutions, particularly in remote or resource-constrained environments.

4.2 CODE:

```
#include <Wire.h>
#include <U8g2lib.h> // U8g2 library for OLED
#include <Arduino.h>
#include <MAX30100_PulseOximeter.h> // MAX30100 library
#include <edge-impulse-sdk/classifier/ei_run_classifier.h>

// Define OLED display (SH1106 128x64)
U8G2_SH1106_128X64_NONAME_F_HW_I2C u8g2(U8G2_R0, /*
reset=/U8X8_PIN_NONE, / clock=/SCL, / data=*/SDA);
```

```

#define NUM_SAMPLES 10
#define RESET_INTERVAL 5000 // Reset interval in milliseconds
int pulse_samples[NUM_SAMPLES]; // Array to store sensor data float displayed_bpm
= 0.0; // Store last classified BPM value
const char* predicted_class = "Unknown"; // Store last predicted status (monitor)
const char* predicted_class_oled = "Unknown"; // Store last predicted status (OLED)
float max_confidence = 0.0; // Store classification confidence
unsigned long lastResetTime = 0;
unsigned long lastUpdate = 0;

// Function to read BPM from MAX30100
void onBeatDetected() {
    Serial.println("Beat detected!");
}

void setup()
{
    Serial.begin(115200);
    u8g2.begin();
    Serial.println("Starting classifier...");

    // Initialize MAX30100
    if (!pox.begin()) {
        Serial.println("FAILED to initialize MAX30100 sensor!");
        while (1);
    } else {
        Serial.println("MAX30100 initialized.");
    }
    pox.setIRLedCurrent(MAX30100_LED_CURR_7_6MA);
    pox.setOnBeatDetectedCallback(onBeatDetected);
}

void loop() {
    unsigned long currentTime = millis();
    pox.update(); // Read sensor data
    displayed_bpm = pox.getHeartRate();

    // Reset system if interval is reached
    if (currentTime - lastResetTime >= RESET_INTERVAL)
    {
        lastResetTime = currentTime;
        Serial.println("System reset.");
    }

    // Update BPM and classification every 4 seconds
    if (currentTime - lastUpdate >= 4000) {
        lastUpdate = currentTime;

        if (displayed_bpm == 0 || displayed_bpm > 220) { // If no heartbeat is detected or

```

invalid value

```
    predicted_class_oled = "Not Detected";
    predicted_class = "Not Detected";
    max_confidence = 0.0;
} else {
    // Collect sensor data
    for (int i = 0; i < NUM_SAMPLES; i++)
    {
        pulse_samples[i] =
        pox.getHeartRate(); delay(100);
    }

    // Convert sensor data into Edge Impulse's required format
    signal_t signal;
    signal.total_length = NUM_SAMPLES;
    signal.get_data = [](size_t offset, size_t length, float *out_ptr) -> int
    {
        for (size_t i = 0; i < length; i++) {
            out_ptr[i] = pulse_samples[offset + i];
        }
        return 0;
    };

    // Run the classifier
    ei_impulse_result_t result = {0};
    int ret = run_classifier(&signal, &result, false);
    if (ret != EI_IMPULSE_OK) {
        Serial.print("ERROR: Inference failed (");
        Serial.print(ret);
        Serial.println(")");
        return;
    }

    // Determine classification based on Edge Impulse model
    float highest_conf = 0.0;
    const char* new_predicted_class = "Unknown";

    for (size_t i = 0; i < EI_CLASSIFIER_LABEL_COUNT; i++)
    {
        if (result.classification[i].value > highest_conf) {
            highest_conf = result.classification[i].value;
            new_predicted_class = result.classification[i].label;
        }
    }

    predicted_class = new_predicted_class;
    max_confidence = highest_conf;

    // Determine classification for OLED based on BPM
    if (displayed_bpm < 60) {
        predicted_class_oled = "Bradycardia";
    }
}
```

```

        else if (displayed_bpm >= 60 && displayed_bpm <= 100)
        { predicted_class_oled = "Normal";
        } else {
            predicted_class_oled = "Tachycardia";
        }
    }

// Display all values at the same time
u8g2.clearBuffer();
u8g2.setFont(u8g2_font_ncenB08_tr);

// Display BPM
u8g2.setCursor(0, 15);
u8g2.print("BPM: ");
if (displayed_bpm == 0 || displayed_bpm > 220)
    { u8g2.print("Not Detected");
    } else {
        u8g2.print(displayed_bpm);
    }

// Display Classification Status
u8g2.setCursor(0, 35);
u8g2.print("Status: ");
u8g2.print(predicted_class_oled);

// Display Model Accuracy
u8g2.setCursor(0, 60);
u8g2.print("Acc: ");
if (max_confidence == 0.0)
    { u8g2.print("--");
    } else {
        u8g2.print(max_confidence * 100, 2);
        u8g2.print("%");
    }

u8g2.sendBuffer(); // Send to OLED

// Serial Monitor Output
Serial.print("BPM: ");
if (displayed_bpm == 0 || displayed_bpm > 220)
    { Serial.println("Not Detected");
    } else {
        Serial.println(displayed_bpm);
    }
Serial.print("Predicted Class: ");
Serial.println(predicted_class);
Serial.print("Model Accuracy: ");
if (max_confidence == 0.0) {

```



```
        Serial.println("--");
    } else {
        Serial.print(max_confidence * 100, 2);
        Serial.println("%");
    }
    Serial.print("Predicted Class (OLED): ");
    Serial.println(predicted_class_oled);
    Serial.println("-----");
}
```

43 SUMMARY

The AI-Based Arrhythmia Detection System, a portable, intelligent, and compact medical device capable of detecting cardiac abnormalities like bradycardia, normal rhythm, and tachycardia in real time, is explained in detail in this chapter. The system is driven by the ESP32-WROOM-32 microcontroller, which is equipped with the MAX30100 pulse oximeter sensor to acquire vital data, i.e., heart rate (BPM). An Edge Impulse-trained machine learning model is embedded within the device, with on-device processing irrespective of cloud services or continuous internet connectivity. The processing of the data is done using Google Colab to ensure reliable and noise-free inputs for model training. After processing and training, the model is put on the ESP32 for inference in real time. The model confidence level, heart condition status, and BPM are clearly shown on a 128x64 OLED display (SH1106) for user engagement. The application of machine learning algorithms for heart health monitoring has resulted in notable progress in the field of biomedical signal processing. Classification algorithms based on electrocardiograms (ECGs) have been more accurate in identifying cardiac irregularities such as arrhythmias. To accurately extract temporal and spatial information from ECG data, recent research has investigated models like Bi-LSTM and 1D-CNN. Real-time application of these methods, including web-based diagnostic tools, has shown great promise.

CHAPTER 5

EXPERIMENTAL RESULTS AND DISCUSSIONS

Smart Heart Rate Classification System was deployed and tested to continuously monitor heart rate in real-time and classify it in real-time. The system employs a MAX30100 pulse sensor, ESP32 microcontroller, and Edge Impulse-trained machine learning model. Results below were obtained during the implementation and evaluation stages.

5.1 Model Architecture

Neural Network settings

Training settings

Number of training cycles ⓘ 50

Use learned optimizer ⓘ ☐

Learning rate ⓘ 0.0005

Training processor ⓘ CPU

Advanced training settings

Neural network architecture

Input layer (1 features)

Dense layer (20 neurons)

Dense layer (10 neurons)

Figure 5.1 Neural Network Settings

From the figure 5.1, The neural network architecture that Edge Impulse uses is referred to as a lightweight architecture aimed at performing real-time embedded classification.

The architecture starts with an input layer that is injected with features extracted from the time-series BPM signal; these include time-domain and frequency-domain features. The extracted features are turned from one or more dense (fully connected) layers with non-linear transformations for learning complex patterns in the input features. Then, a dropout layer is applied to promote generalization and reduce overfitting by randomly dropping a certain percentage of neurons during the training epochs. The final layer of the architecture is the SoftMax output layer, which gives the final output mapped to a probability distribution over the three classes for classification: bradycardia, normal, and tachycardia.

5.2 Machine Learning Model Deployment and Testing (Edge Impulse)

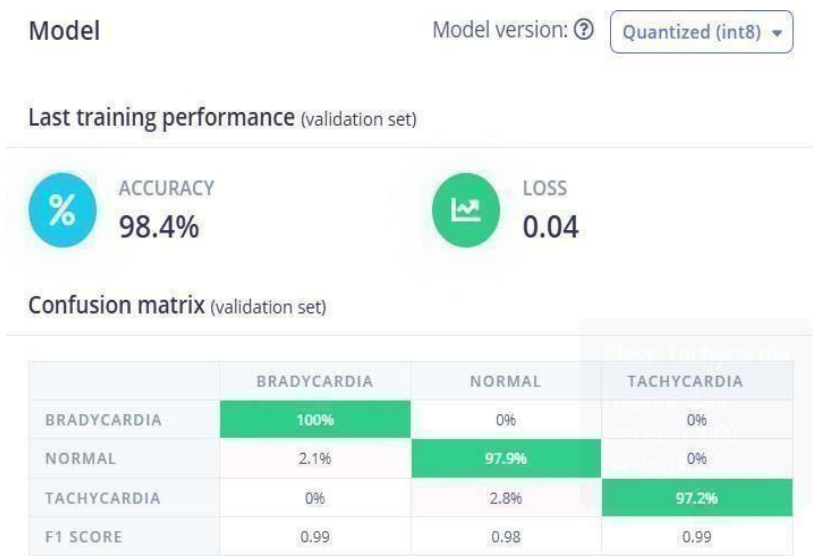


Figure 5.2 Machine Learning Model Deployment and Testing

The machine learning model was quantified (int8) and ran on Edge Impulse. Performance on the validation set was excellent, with accuracy of 98.4% and loss of 0.04. The confusion matrix also confirms high classification accuracy

for all three classes: Bradycardia (100%), Normal (97.9%), and Tachycardia (97.2%). The F1 scores of all classes are also high, ranging from 0.98 to 0.99, establishing a well-balanced model. The model is now edge deployment ready, correctly classifying real-time health conditions on low-power devices without the loss of accuracy.

5.3 VSCode Terminal Output Observation

```
--- Classification Results ---  
Bradycardia: 0.39%  
Normal: 99.61%  
Tachycardia: 0.00%  
Predicted Class: Normal  
Model Accuracy: 99.61%  
System reset.  
  
--- Classification Results ---  
Bradycardia: 50.00%  
Normal: 50.00%  
Tachycardia: 0.00%  
Predicted Class: Bradycardia  
Model Accuracy: 50.00%  
  
--- Classification Results ---  
Bradycardia: 0.00%  
Normal: 0.00%  
Tachycardia: 99.61%  
Predicted Class: Tachycardia  
Model Accuracy: 99.61%
```

Figure 5.3 VS code terminal output for prediction

```
---Classification Results ---  
-Bradycardia: 0.39%  
Normal: 99.61%  
Tachycardia: 0.00%  
Predicted Class: Normal  
Model Accuracy: 99.61%
```

---Classification Results ---

-Bradycardia: 50.00%

Normal: 50.00%

Tachycardia: 0.00%

PredictedClass:Bradycardia

Model Accuracy: 50.00%

---Classification Results ---

-Bradycardia: 0.00%

Normal: 0.00%

Tachycardia: 99.61%

PredictedClass:Tachycardia

Model Accuracy: 99.61%

5.4 DataExplorer and On-device Performance (Edge Impulse)

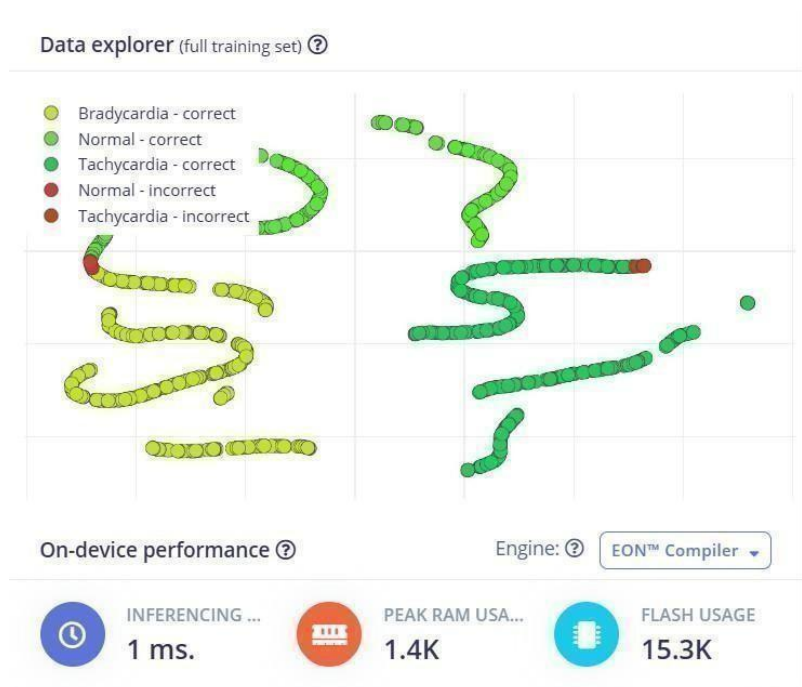


Figure5.4 Data Explorer and On-device Performance

From the figure5.4, This plot illustrates the Data Explorer in Edge Impulse, where every colored dot represents a data sample correctly or incorrectly labeled by the model. Most of the samples are correctly labeled as bradycardia, normal, or tachycardia, indicating high model accuracy with minimal misclassification. The following on-device performance measures demonstrate outstanding efficiency with:

Inferencing time: 1 ms

MaximumRAMutilization: 1.4 KB

Flash usage: 15.3 KB This is done to ensure the model remains lean and optimized for embedded hardware to allow real-time classification without overloading system resources.

5.5 Hardware Setup and Implementation

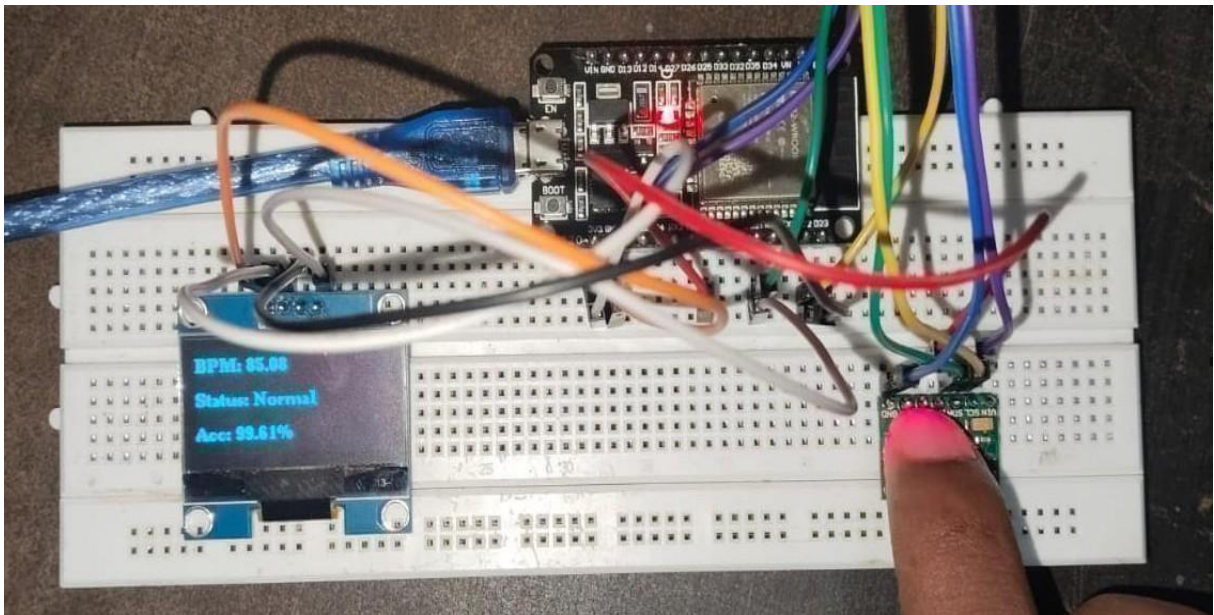


Figure 5.5 Hardware Setup of the Proposed System

From the figure5.5, The hardware prototype of the suggested system consists of an ESP32- WROOM-32 microcontroller interfaced with a MAX30100 pulse

sensor module through proper GPIO pins. The MAX30100 sensor is used to measure the pulse rate of the user by sensing the variation in blood volume. The connections were done on a breadboard setup for convenient testing and modifications. The ESP32 processes the sensor readings and sends them to the Edge Impulse model running on-device. The entire system is powered through the ESP32's USB interface, connected to a laptop running VS Code with platform IO extension. This setup ensures portability, ease of integration, and suitability for personal and clinical health monitoring environments.

5.6 Limitations

No graphical user interface (GUI) or cloud dashboard was integrated in this version. Real-time monitoring is limited to serial terminal output without mobile/web alerts. The dataset size for model training was moderate, which might affect accuracy for edge-case BPM values. The system's performance establishes its potential for future enhancements like integration with cloud dashboards, mobile apps, or wearable designs for personal health tracking.

5.7 SUMMARY

In brief, Smart Heart Rate Classification System is a smart and efficient solution for real-time heart rate classification and monitoring in healthcare monitoring systems. Pulse sensor integration with ESP32 microcontroller and Edge Impulse-trained AI model ensures accurate, real-time heart rate analysis. Terminal-based monitoring and VS Code and Edge Impulse-based automated classification features ensure real-time feedback of the heart condition of the patient, i.e., Bradycardia, Normal, and Tachycardia detection. Early identification of abnormal heart rate patterns ensures early intervention and enhanced patient safety. Cloud-connected data logging and Edge Impulse's advanced analytics feature ensure trend analysis and future predictive health diagnostics. The lightweight implementation of the system also ensures a sustainable and viable solution for portable and remote healthcare installations. Future extensions involve AI-based anomaly prediction models, adaptive learning algorithms for dynamic threshold adjustment based on personal health profiles, and Internet of Medical Things (IoMT) platform integration for networked, holistic health systems. Extensions will also enhance the system's capability in personalized and preventive healthcare applications.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The AI-Enhanced Arrhythmia Detection System has been effectively deployed to provide a real-time, intelligent solution for heart rate monitoring and classification using embedded machine learning. Based on the ESP32 WROOM 32 microcontroller and the MAX30100 pulse sensor, the system continuously monitors heart rate data and accurately classifies it into bradycardia, normal, or tachycardia conditions. The machine learning model, trained on the MIT-BIH dataset using Edge Impulse, has been demonstrated to have high accuracy and low latency, making it highly appropriate for deployment on edge devices without the need for external servers. The overall process—from data collection and preprocessing in Google Colab to model training, deployment using Platform IO, and real-time inference—was optimized for efficient integration. The terminal output demonstrates the system's effectiveness in detecting transitions between heart rate categories, providing immediate feedback to the user. This embedded solution offers a portable, low-cost, and scalable solution for early detection of heart anomalies, which is highly beneficial in remote and resource-poor healthcare settings. The system offers a robust platform for future enhancements such as adding cloud-based alerts, adding more sensor inputs, and adding adaptive learning for continuous performance improvement.

6.2 FUTURE ENHANCEMENT

In the future, the AI-Enhanced Arrhythmia Detection System can be further expanded with real-time cloud connectivity for remote health monitoring, allowing doctors and caregivers to view patient data in real-time remotely. Support for mobile apps can further improve usability, providing users with graphical reports, health reports, and emergency alerts. Support for wearable form factors like smartwatches or fitness bands would further improve convenience and accessibility for ongoing monitoring. Advanced machine learning algorithms like adaptive learning algorithms can be employed to further improve prediction accuracy over time by adapting to new user-specific data. Increasing the number of biosensors like ECG leads, temperature, or blood pressure sensors would further allow more thorough analysis of cardiovascular health. Lastly, support for medical data security standards like HIPAA and GDPR will be required to provide secure data handling and user privacy in real-world medical use.

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ANNEXURE I
BASE PAPER



An improved hybrid AI model for prediction of arrhythmia using ECG signals

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ARTICLE INFO

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ABSTRACT

An arrhythmia is a disorder where the heart beats out of its usual rhythm. It means the heart beats irregularly. The rate of a person's heartbeat is determined by electrical signals flowing through the heart. These signals are produced by the Sino Atrial (SA) node in the heart which acts as a natural pacemaker. If these electrical signals are irregular, it leads to heart problem. In recent years, the individuals who are diagnosed with this condition has increased. In this paper we suggested a hybrid model for feature extraction and classification. Many deep learning algorithms and machine learning algorithms are used, from that two layered BiLSTM gives the best accuracy for feature extraction. Random forest gives the best accuracy for classification. Also feature reduction technique is used to minimize the training and testing time. The proposed hybrid model which comprised of BiLSTM and Random forest gives the best accuracy of 98.84% and shows the sensitivity and specificity of above 98% for all the classes. The proposed hybrid model with feature reduction technique produces an accuracy of 98.46% with lesser prediction time of 0.089 s. The hybrid BiLSTM + Random Forest + PCA model generates a higher average accuracy of 98.30% compared to LSTM + Random Forest + PCA hybrid model with 10-fold cross validation, which again ensures the consistency of the proposed models.

1. Introduction

Cardio vascular diseases are one of the fast-increasing non-transmissible ailments in modern times. Arrhythmias are one of the most common heart ailments connected with heart beat cycles. An arrhythmia is a disorder where the heart beats out of its usual rhythm. It means the heart beats irregularly. A heart beat cycle is determined by the electrical impulses that passes through heart myocardial tissue. These electrical impulses determine the contraction and relaxation of various portions of the myocardial tissue thus giving rise to the heart beat rhythm. A healthy human heart has a regular identifiable heartbeat pattern. A heartbeat rhythm of a person with arrhythmia will be irregular. This rhythm may be faster or slower depending upon electrical activity, free flow of blood in the heart, myocardial tissue strength and damage in heart muscle among other factors.

The human heart is made up of two upper and two lower chambers through which the blood flows. Right atrium is in the upper right part of the heart which receives the de-oxygenated blood inflow from other parts of the body. The right atrium then contracts and the blood flow into the right ventricle which is situated directly below the right atrium. The right ventricle then contracts and pushes the blood into the lungs where the blood gets oxygenated. The blood from the lungs is received by the left atrium located in the heart's upper left part. This part then contracts and pumps the blood into the left ventricle located below the left atrium. The left ventricle then contracts and in doing so

pumps the oxygenated blood for other parts of the body. One important characteristic in these contractions is both of the atriums will contract simultaneously while sending blood to their respective destinations and then both of the ventricles will contract simultaneously sending blood to their respective destinations.

These above-mentioned contractions in the atriums and ventricles of the heart are triggered by the electrical impulses generated in two important nodes located in the right atrium. These impulses determine which part of the heart to contract and thus the rhythm of the heart. The first step in the heart beat cycle involves the sinoatrial (SA) node. This node acts as the natural pacemaker. An electrical impulse is generated in the SA node and propagates to the both atriums and stimulates them to contract. This electrical signal is received by the Atrioventricular (AV) node situated on the other side of the right atrium close to the AV valve. There is a small delay in passage of electrical impulse to the ventricles to correctly ensure that all the blood from atrium flows into the ventricles before the start of ventricular contraction. The AV node then passes the electrical signals to the AV bundle which reaches the bottom of the heart. Then these electrical signals propagate upwards to the muscles of the heart. All these activities spread to the rest of the heart and can be recorded by attaching electrical leads on the surface of the body through the ECG instrument. The ECG records the heartbeats which contains atrial contraction and contraction or depolarization of ventricles.

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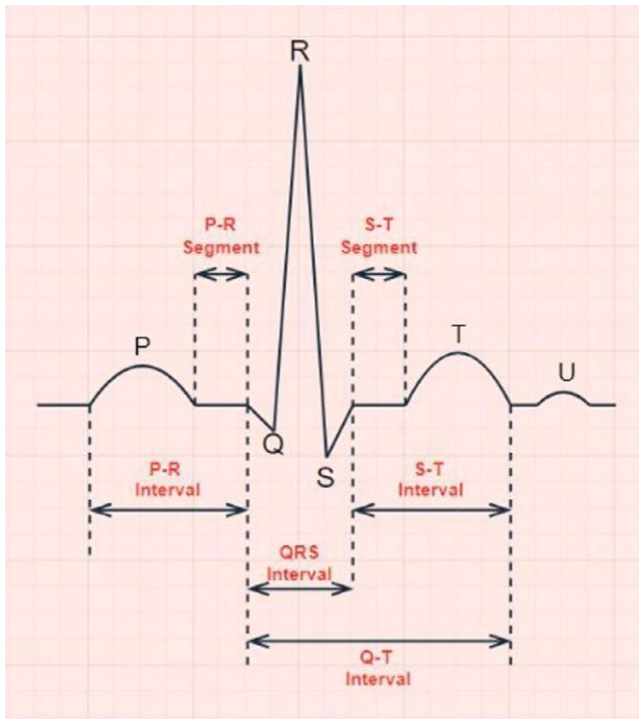


Fig. 1. ECG signal.

The ECG signal shown in Fig. 1, produces an identifiable wave for each heartbeat cycle. Each segment of the wave represents a certain activity in the heartbeat cycle. When the blood sufficiently fills the atrium the SA nodes trigger an electrical pulse which causes atrial depolarization represented in the ECG signal by P wave. The P wave is followed by 100 ms of atrial contraction. The transmission of electrical signal from the SA node to the AV node is represented by the PQ section. The QRS segment denotes the firing of AV node or ventricular depolarization. The Q waves indicate the pulses flowing through the bottom of the heart. The R wave is generated when the electrical signal flows in the sides of the ventricles. The S wave denotes the last phase of ventricular depolarization at the bottom heart muscles. Also in the same time atrial repolarization happens but the signal is hidden behind the large QRS complex. The ST segments denote the maintenance of ventricle contraction. The wave represents the ventricular repolarization which concludes one cycle of the heartbeat.

By using ECG signals the cardiologist can identify different types of cardiac arrhythmia through the variations in the waves, segments and intervals. Doctors can quickly detect the presence of arrhythmia in the ECG data by applying machine learning and deep learning models for automated feature extraction and categorization of various forms of arrhythmia.

The topics which are discussed in this paper are given below. In Section 2, the works of other researches based on feature extraction and classification are given. In Section 3, the proposed architecture with feature extraction, feature reduction and classification is discussed. The experimental findings of the different algorithms are discussed in Section 4. Finally, in Section 5 the conclusion of the proposed work is given.

2. Literature Survey

There are lot of works done previously based on different algorithms for automatic diagnosis of arrhythmia by researches. The important features in ECG which are used to differentiate the presents of various

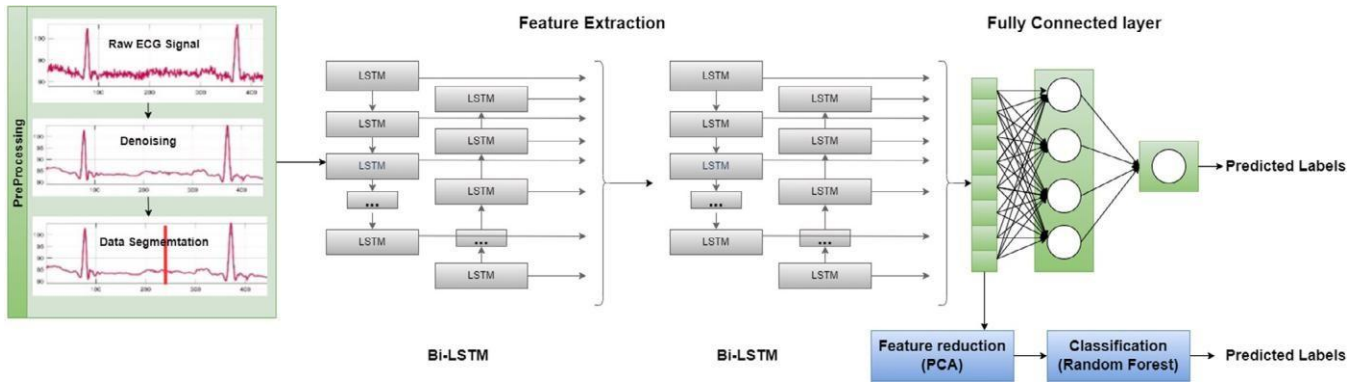
type of arrhythmia are the morphological features and segment features. [1] developed a system to identify the abnormal heartbeat from the compressed ECG. The unwanted noises in the signals are removed by Wavelet transform, Discrete Cosine Transform (DCT) is used to extract ECG parameters and random forest is used for the classification. A new approach is proposed by [2] which uses discrete orthogonal stockwell transform to identify time frequency parameters. These parameters are reduced by PCA dimensionality reduction method. The reduced dimensions are given as input to the SVM algorithm. [3] used Least Mean Squares (LMS) adaptive filtering to denoise the signal and Heart rate variability (HRV) data are extracted from the ECG signals by Discrete Wavelet Transform (DWT). After the feature extraction SVM classifier and KNN classifiers are used for Classification. From that SVM classifier gives better performance.

Discrete Wavelet Transforms (DWT), Autoregressive Modeling (AM), Continuous Wavelet Transform (CWT) and Maximum Overlap Discrete Transform (MODWT) are used by [4] for the purpose of feature extraction and Convolutional Neural Networks (CNN), naive Bayes, quadratic discriminant, support vector machine (SVM) and k-nearest neighbor algorithms are used for classification. The accuracy of 99% is produced by KNN and quadratic discriminant algorithms. A new feature extraction algorithm is suggested by [5] based on clustering by combining morphological features. The morphological features such as Amplitude, duration and intervals are collected from the ECG signals. For classification of extracted features three classification algorithm such as KNN, neural network and SVM are used. They got the best accuracy for KNN algorithm which is 97.70%.

A division of machine learning is deep learning. Deep learning offers the benefit of automatically extracting information from the ECG. A method for identifying and categorizing shockable and non-shockable rhythms was put out in [6]. To create the Time-frequency maps, the Continuous Wavelet Transform (CWT) is employed, then Convolutional Neural Networks (CNN). To identify arrhythmia using single lead brief electrocardiogram (ECG) records [7] suggests Multiscaled fusion of deep convolutional neural network (MS-CNN). Two streams of convolutional networks are employed, each with a distinct filter size, to capture various scale information.

The convolutional neural network (CNN) model is used by [8], also uses the Two-dimensional Short-time Fourier Transform (2D STFT) and Continuous Wavelet Transform (CWT) as a filter to obtain information about frequency variations. Feature extraction and classification are performed using the CNN architecture. 1D and 2D CNN models are used to compare the result. 99% recall rate and 97% accuracy rate are achieved by the proposed work and 60.5% and 56% accuracy rates are achieved by 1D and 2D CNN models. To identify the presents of Atrial Fibrillation (AF) by finding the premature atrial complexes (PACs) [9] used ResNet and Wide-ResNet CNN models. To minimize the overfitting problem, they proposed a method by combining sensitivity and orthogonality penalty term. Resampling is also done to overcome the problem of imbalanced data. For ECG classification, [10] used a novel end-to-end deep multi-scale fusion CNN model. For feature extraction, multiple convolution kernels are employed.

Since there may be lags of uncertain duration between significant occurrences in a series data, LSTM networks are well-suited for the classification, processing, and prediction of time series data. [11] collected the linear and nonlinear features from the sensors connected with Internet of Things. For the feature extraction and classification of arrhythmia intervals and normal sinus intervals they have used a combination of both deep convolutional neural network (CNN) and a long short-term memory (LSTM). They also performed 10-fold cross validation, and got the average accuracy, sensitivity, and specificity of 99.6%, 98.29% and 99.73%. Two deep learning methods are used by [12] to extract the features. Bidirectional LSTM is utilized to learn the global features, while the deep residual network is utilized to extract local features. This model gives the overall f1- score of 0.806.



/fig. 2. Proposed hybrid architecture.

Multifractal parameters are also used to classify arrhythmia, [13] employed feature-based bidirectional LSTM (bi-LSTM). Suitable features are extracted using multifractal detrended fluctuation analysis. Nonlinear, stochastic, and complex fluctuations of ECG signals are studied followed by sending it to a single layer bi-LSTM model. The ECG classification method must be efficient and lightweight for wearable technology that provide constant monitoring. So, [14] proposed an algorithm which extracts RR interval and wavelet information from signals. The extracted features are trained using two RNN architectures. The output obtained from the two RNN models are blended using Multi-level Perceptron (MLP) algorithm to predict the final output. The various algorithms used in previous works for the classification of arrhythmia are K-nearest neighbor (KNN) algorithm [15,16], random forest [17], Decision tree [18] and Support Vector Machine (SVM) [19].

3. Proposed system

For the automatic feature extraction and classification of arrhythmia we have proposed a hybrid model which includes preprocessing to denoise the ECG signals, data segmentation to separate each beat, feature extraction to learn features, feature reduction to minimize dimensions followed by classification. Fig. 2 depicts the proposed hybrid architecture.

3.1. Dataset description

In this study, Massachusetts Institute of Technology-Beth Israel Hospital (MIT-BIH) Arrhythmia Dataset [20] is used. Dataset contains the ECG signals taken from 47 persons recorded for 24 h. This data set contains ECG signals of subjects aged between 23 and 89 and also contain 53% male and 47% female cardio signals. The data are provided in two files. The .csv files contain ECG signals and the .txt files contain the annotations for the ECG signals. In this work we are considering the data for five classes of arrhythmia. Non-ectopic beats (normal beat), Supraventricular ectopic beats (SVE) which occurs when there is a problem in upper chambers of the heart, Ventricular ectopic beats (VEB) which occurs when there is a problem in lower chambers of the heart, Fusion Beats which occurs when there is a problem in both upper and lower chambers of the heart (combination of SVE and VEB) and Unknown Beats.

3.2. PreProcessing

Various preprocessing techniques such as denoising, segmentation and sampling are used in our proposed model which are discussed below one by one.

Table 1

Number of samples in each class.

Classes	Number of samples
Normal	75011
SVE	2546
VEB	7129
Fusion	802
Unknown	982

3.2.1. Denoising

Denoising is performed to minimize the noise or unwanted data from the ECG signal. For denoising Discrete Wavelet Transform (DWT) is used. Three steps are involved in DWT. The first step involves the forward transformation of ECG signal to wavelet domain. The second step involves the choice of the wavelet type and choosing a threshold and applying in the wavelet coefficient are three important characteristics in denoising. Several wavelet transformation classes are available. Out of which 22 orthonormal transformations are frequently used. They consist of 7 symmlets, 9 daubelets, 1 haar and 5 coiflets. The wavelet transform class used in this work is symmlets and the threshold value is fixed at 0.04. The equation for discrete wavelet transform of the sequence $x_i[n]$ where $i = 0, 1, \dots, n$ is given as in Eq. (1)

$$y_i[k] = \sum X[n]g_i[2k - n] \quad (1)$$

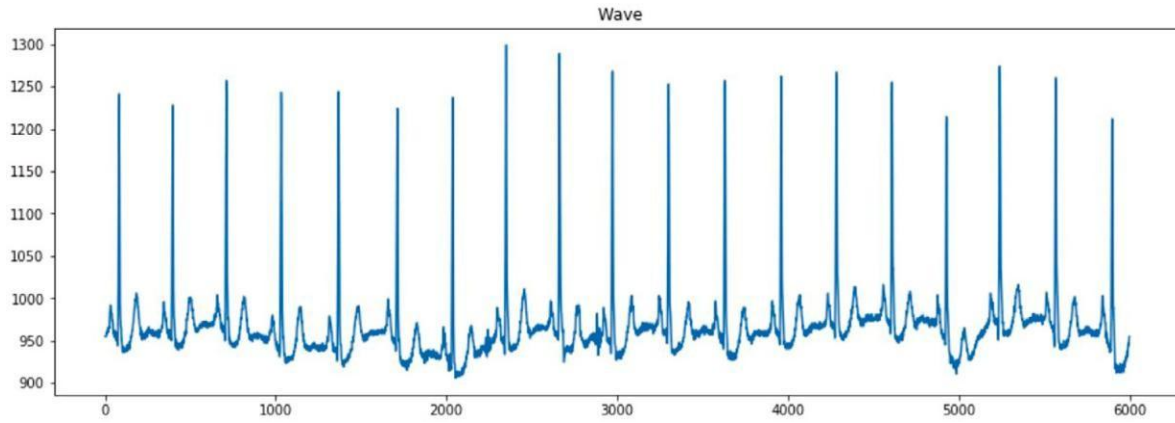
where $y_i[k]$ is the discrete wavelet sequence and $g_i[2k - n]$ is the basis sequence.

3.2.2. Segmentation

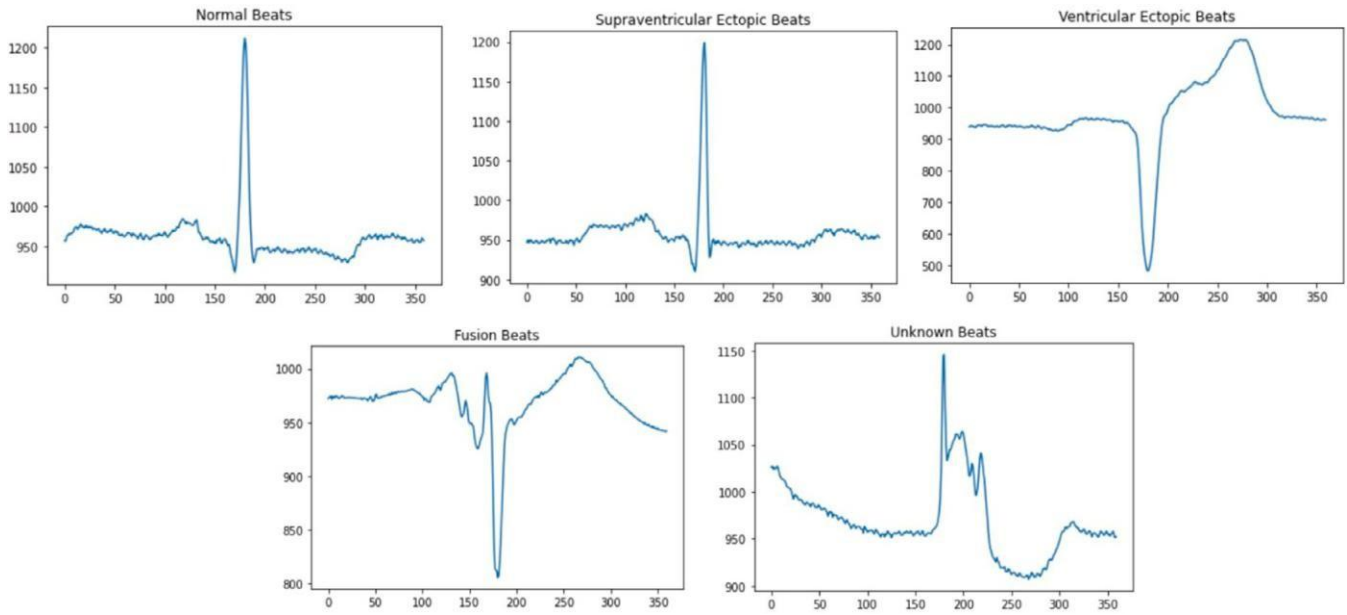
The dataset contains the ECG signal with continuous beats. Fig. 3 shows the continuous ECG signals given in the dataset. Segmentation is used to separate a single beat from the continuous beats. Each beat contains QRS complex. There are several ways to detect QRS complexes. The doctors used to find the R-peaks and variations from QRS complex using electrocardiogram signals with lot of practice. Machine learning did the detection by using algorithms. In this study, segmentation is accomplished by fixing the window size. The R-peaks' positions are given in the dataset. The size of the window is set at 180 features. A single beat is defined as the size of the window before and after the R-peak. The number of instances of different classes after segmentation are shown in Table 1. Figs. 4 and 5 shows the segmented signals before denoising and after denoising for each class.

3.2.3. Resampling

Resampling is the process of balancing an unbalanced dataset. The dataset has a varied number of data for each class. When compared to the class with the most instances, the number of instances of the class with the fewest instances is less than one percent. To balance the dataset, resampling is performed. The number of rows desired for each



/fig. 3. ECG signal given in the dataset.



/fig. 4. Segmented ECG signals before denoise.

class is specified as input. If the number of instances of the class exceeds the specified input, the method removes a few instances at random. If the number of instances is less than the provided input, the algorithm replicates the random rows to make the class equal in size. Following sampling, each class considered has 5000 samples.

3.3. Feature extraction

After the resampling of data, the dataset is splitted into training and testing. 80 percent of the data are allocated for training and 20 percent of the data are allocated for testing. As a result, there are 5000 instances in the testing and 20000 in the training set. The feature extraction is used to learn the signal features. Long Short Term Memory (LSTM), RNN GRU and bidirectional LSTM are the different techniques used in this work. Below is a discussion of these various algorithms.

3.3.1. LSTM

Long Short-Term Memory (LSTM) has the ability to learn dependencies in the time series data. This algorithm is created to solve the long-term dependency issue. LSTM comprises of four key components. There are three gates and a cell. The functionality of the gate is to control the

flow of data to the cell and the functionality of the cell is to hold the values in random time. The equation for LSTM is given below.

$$F(T) = \sigma_g(\omega_F * x(T) + \alpha_F * h(T-1) + \beta_F) \quad (2)$$

$$I(T) = \sigma_g(\omega_I * x(T) + \alpha_I * h(T-1) + \beta_I) \quad (3)$$

$$O(T) = \sigma_g(\omega_O * x(T) + \alpha_O * h(T-1) + \beta_O) \quad (4)$$

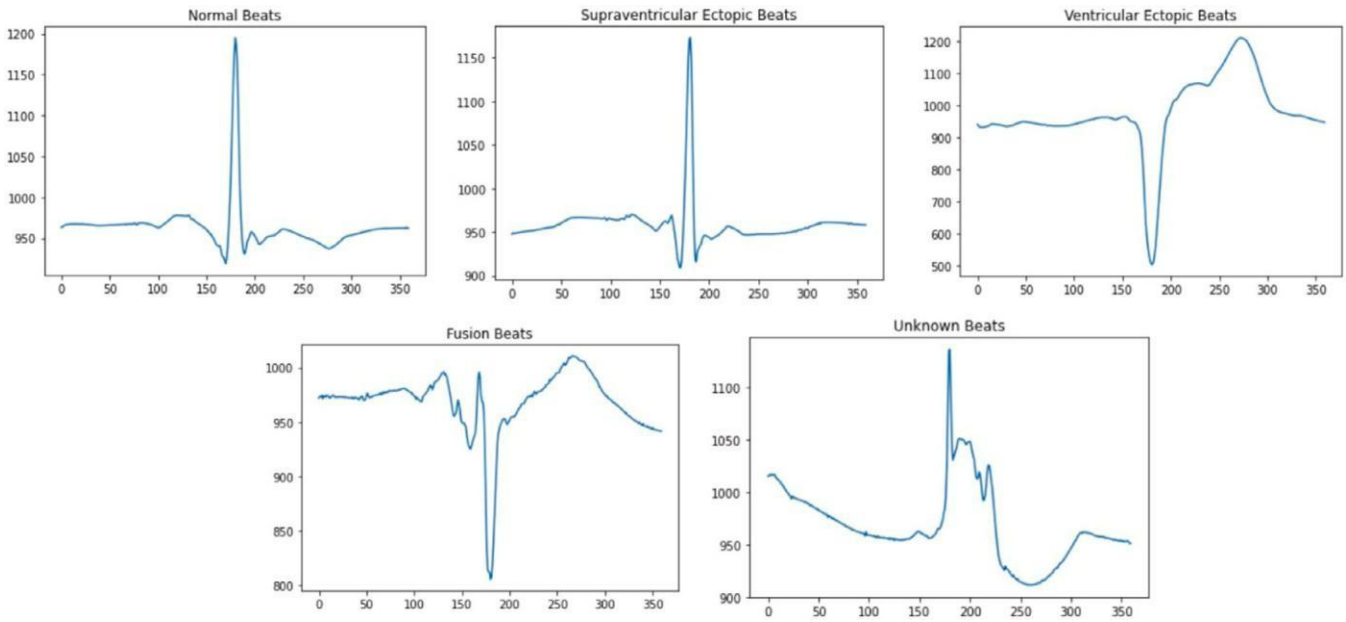
$$C'(T) = \sigma_g(\omega_C * x(T) + \alpha_C * h(T-1) + \beta_C) \quad (5)$$

$$C(T) = F(T).C(T-1) + I(T).C'(T) \quad (6)$$

The Eq. (2) is the equation for forget gate, Eq. (3) is the equation for input gate, Eq. (4) is for output gate and Eq. (5) represents cell state. In the above equations, $\omega_F, \omega_I, \omega_O, \omega_C$ are weight matrices, $\beta_F, \beta_I, \beta_O, \beta_C$ are biases and $h(T-1)$ is previous hidden state.

3.3.2. RNN GRU

Gated recurrent units (GRUs) is created to overcome vanishing gradient problem which comes in Recurrent Neural Network (RNN). This algorithm is related to LSTM which includes forget gate, but it has



/fig. 5. Segmented ECG signals after denoising.

less parameters compared to LSTM and it does not contain the output gates. The other gates in GRU are update gate which determines the number of data required to pass to future state from the previous state. Eq. (7) shows the formula for update gate.

$$U(T) = \sigma(\omega(U) * x(T) + \alpha(U) * h(T-1)) \quad (7)$$

here $\omega(U)$ and $\alpha(U)$ are weights, hidden information from prior state is represented as $h(T-1)$. The Reset gate which determines how much previous data to forget. Eq. (8) is the formula for reset gate.

$$R(T) = \sigma(\omega(R) * x(T) + \alpha(R) * h(T-1)) \quad (8)$$

where $\omega(R)$ and $\alpha(R)$ are weight parameters.

3.3.3. Bidirectional LSTM

Bidirectional long-term dependencies are found by using BiLSTM algorithm. BiLSTM algorithms are used mostly in sequence or timeseries data. The found dependencies are helpful when the application wants to remember the full timeseries information at each point. BiLSTM contains two LSTM layers among that one is used for forward learning and another one is used for backward learning. Eqs. (9) and (10) stand for the hidden states of forward and backward learning, respectively.

$$h(T) = LST M(x, h(T-1)) \quad (9)$$

$$\bar{h}(T) = LST M(x, \bar{h}(T+1)) \quad (10)$$

The BiLSTM output is computed using Eq. (12).

$$y(T) = \sigma(\omega[h(T), \bar{h}(T)] + \beta) \quad (11)$$

Where ω represents bias and β represents weight matrix. This algorithm increases the number of data for the small datasets thus shows high accuracy.

3.4. Feature reduction

Feature reduction is done to minimize the features after feature extraction and to reduce the training and testing time. The features are extracted from the flatten layer of the deep learning model. In the feature extraction phase, the LSTM algorithm extracts 3600 features for each beat while the BiLSTM method extracts 7200 features for each beat. The two feature reduction techniques used are Linear Discriminant Analysis (LDA) and Principal Component Analysis (PCA), and they are detailed below.

3.4.1. Principal Component Analysis (PCA)

By using PCA, data with more number of features are projected to a new space with less dimensions. PCA finds the covariance among the extracted features to find the pattern in the data and it can be calculated using the Eq. (12).

$$Cov(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad (12)$$

Then it calculates the eigenvectors and eigenvalues from the covariance matrix. k largest eigen vectors are chosen by arranging the eigenvalues corresponding to them in decreasing order where k is the new dimension. This algorithm creates the projection matrix using chosen eigenvectors. Finally PCA converts the d-dimension features to k-dimension features with the help of projection matrix. After the feature extraction technique using LSTM, 3600 features are reduced to 150 features and 7200 features after BiLSTM are reduced to 150 features by using PCA feature reduction technique.

3.4.2. Linear Discriminant Analysis (LDA)

The features in our data are reduced using LDA, a multiclass classification technique. This algorithm tries to identify the subspace for features which increases the class separability. Linear Discriminant Analysis finds the mean values of the data in all the classes and variance to predict the class by using Gaussian distribution. The Eq. (13) is used to compute the mean of the data after projection.

$$\bar{y} = \frac{1}{\sum_{i=1}^n} = \quad (13)$$

where n represents the number of instances in class c, $\bar{y}$ represents the

point and its projection is shown as $\bar{y}$. LDA reduces features to c-1 features, while c is the class label. Because there are 5 classes in this work, 3600 features after LSTM and 7200 after BiLSTM are reduced to 4 features.

3.5. Classification

The reduced features are given as input to various machine learning methods. The following techniques are used for classification: Decision tree, Random forest, Naive Bayes, KNN, SVM with Polynomial kernel, SVM with rbf kernel and SVM with Sigmoid kernel. These algorithms are discussed below.

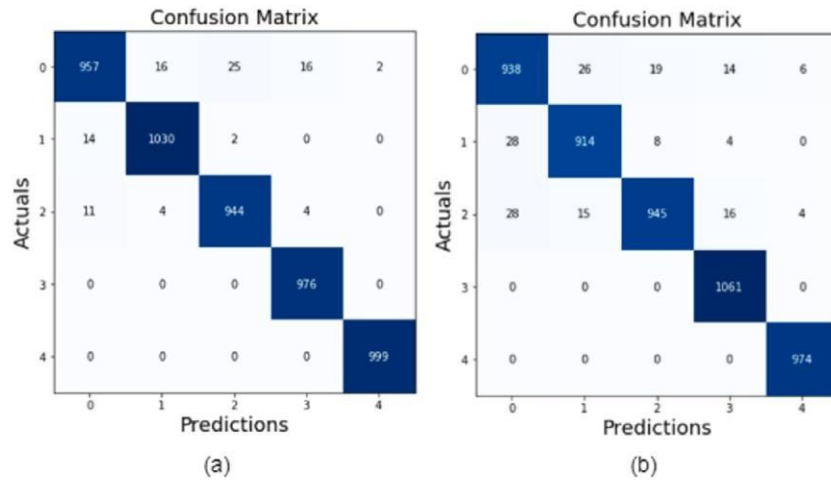


Fig. 6. Confusion matrix (a) LSTM with PCA + Random forest (b) LSTM with LDA + Random forest.

3.5.1. k-nearest neighbor

The k closest neighbors are found using this approach nevertheless of target class. The number of nearest neighbors to be considered is passed as the parameter and represented as k. The distance between the new input and the trained points are calculated. The minimum distanced k trained points are considered and the class of that points are counted. The class with maximum number of points close to the new point is given as the output. Euclidean distance is used by KNN to find the distance between points. The formula to find distance is given in Eq. (14).

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (14)$$

In the above equation x indicates the new input point, y indicates the trained point and the number of features is given as n .

3.5.2. Naive Bayes

The main considerations of this classifier algorithm are features that are independent and equal. This algorithm works based on conditional probability. The probabilities of all the classes are calculated when new input is given. The class with the high probability is considered as the output. Even with small amounts of data and with a greater number of features, this method produces the best results. The probability equation is given in Eq. (15).

$$P(X|Y) = P(Y|X) * P(X)P(Y) \quad (15)$$

where $P(X|Y)$ indicates the posterior probability.

3.5.3. SVM

SVM classifier divides n-dimensional space to different labels by using boundary. The extreme points, otherwise known as support vectors are used to generate the decision boundaries. Position of new input point in the n-dimensional space determines output class of the new point. For both regression and classification, the SVM classifier can be used. It is difficult to construct the hyperplane or the decision boundary for non-linear data or high dimensional data using SVM. Kernel functions are used in SVM to solve this issue by linear classifier. The different SVM kernel used are Sigmoid kernel, Polynomial kernel and Gaussian radial basis function kernel.

$$K(x, y) = (x \cdot y + 1) \quad (16)$$

$$K(X, X) = \exp(-r \|X - X\|^2) \quad (17)$$

$$K(x, y) = \tanh\left(\frac{x \cdot y + c}{\sqrt{1 + c^2}}\right) \quad (18)$$

Eq. (16) is the equation of Polynomial kernel, where the degree is given as r . Eq. (17) is the formula for Gaussian radial basic functional kernel and Eq. (18) is the equation of Sigmoid kernel, where the kernel parameters are c and r .

3.5.4. Decision tree

This algorithm divides the features in the dataset to form tree. A decision tree includes the root node, decision node and leaf node. Entropy and Information gain for all the features are calculated. The formula to calculate Entropy and information gain are given in Eqs. (19) and (20).

$$E = -\sum_{i=1}^n f_i \log_2 f_i \quad (19)$$

$$\text{gain}(d, x) = E(d) - E(d, x) \quad (20)$$

where n is the number of samples of label i in a node, $E(d)$ denotes the number of distinct labels, $E(d, x)$ represents target label, f_i represents the attribute to be divided on. The attribute which has high Information gain or low Entropy is selected as the root node. This calculation is made recursively for subdata to construct decision tree. The leaf node indicates the output of the new data.

3.5.5. Random forest

Many decision trees are created using this algorithm. This algorithm selects n random samples from the dataset. For each sample, random forest constructs a separate decision tree. In random forest each decision tree uses gini index to split the features into trees. The equation for gini index is given in Eq. (21).

$$\text{gini} = 1 - \sum_{i=1}^n (f_i)^2 \quad (21)$$

f_i represents number of samples of label i in a node, $E(d)$ denotes the distinct labels. All decision trees predicts the output. The final prediction is based on majority voting. The result of the random forest algorithm is the class with the most votes. If the number of decision trees are more, then the accuracy will be high.

4. Result and discussion

The dataset used for this paper is MIT-BIH dataset. There are totally 86,470 samples distributed unevenly across five classes. After data sampling, there are 25000 samples in total, spread evenly among five classes. There are 5000 samples in each of the five classes. We have implemented the feature extraction, feature reduction and classification modules with denoising using different algorithms to compare the

Table 2
Results of LSTM with Other ML algorithms.

Methods	Classes	Sensitivity	Specificity	F1-Score	Accuracy	Testing time
KNN	Normal	0.91	0.96	0.93	95.62%	340.34 S
	SVE	0.95	0.9	0.95		
	VEB	0.93	0.96	0.96		
	Fusion	0.98	0.94	0.95		
	Unknown	0.99	0.94	0.98		
Naive Bayes	Normal	0.40	0.83	0.54	74.52%	0.63 S
	SVE	0.72	0.74	0.73		
	VEB	0.79	0.73	0.80		
	Fusion	0.86	0.71	0.86		
	Unknown	0.93	0.69	0.75		
SVM with Polynomial kernel	Normal	0.79	0.90	0.83	87.98%	238.30 S
	SVE	0.77	0.90	0.87		
	VEB	0.91	0.87	0.95		
	Fusion	0.92	0.86	0.95		
	Unknown	0.99	0.85	0.82		
SVM with Radial basis function	Normal	0.94	0.96	0.92	96.06%	119.17 S
	SVE	0.93	0.96	0.96		
	VEB	0.96	0.95	0.97		
	Fusion	0.96	0.95	0.96		
	Unknown	0.99	0.95	0.99		
SVM with Sigmoid kernel	Normal	0.75	0.77	0.73	77.34%	171.47 S
	SVE	0.74	0.78	0.80		
	VEB	0.72	0.78	0.72		
	Fusion	0.76	0.77	0.74		
	Unknown	0.87	0.74	0.89		
Decision tree	Normal	0.88	0.97	0.91	95.78%	0.016 S
	SVE	0.96	0.95	0.95		
	VEB	0.95	0.95	0.95		
	Fusion	0.99	0.94	0.98		
	Unknown	0.99	0.94	0.99		
Random forest	Normal	0.96	0.99	0.97	98.58%	0.175 S
	SVE	0.98	0.98	0.99		
	VEB	0.98	0.98	0.99		
	Fusion	0.99	0.98	0.99		
	Unknown	0.99	0.98	0.1		

performance. The LSTM model gives the accuracy of 94%, GRU model gives the accuracy of 85% and BiLSTM model gives the accuracy of 96% when used for both feature extraction and classification. When multi-tier algorithm is used by combining the other machine learning algorithms for classification the accuracy is increased.

Table 2 shows the result of LSTM with other machine learning algorithms along with the testing time. Among all the algorithms Random forest shows better accuracy of 98.58%. Table 4 shows the result of BiLSTM with other machine learning algorithms along with the testing time. Among all the algorithms Random forest shows better accuracy of 98.84% with the sensitivity, specificity and f1-score of above 98% for all the classes.

To reduce the prediction time feature reduction techniques are used. LSTM extracts 3600 features from each ECG signal. These features are reduced by using PCA and LDA. PCA takes 5.848 s and LDA takes 72.722 s for feature reduction while using LSTM as the feature extraction technique. Table 3 shows the result of LSTM with other ML algorithms after the feature reduction techniques PCA and LDA techniques. From this table Random forest with PCA and Random forest with LDA gives better accuracy of 98.2% and 97.24%. Fig. 6 shows the confusion matrix of LSTM, Random forest with PCA and LSTM, Random forest with LDA. As LDA takes more time for feature reduction, Random forest with PCA performs better and shows best accuracy and prediction time.

BiLSTM extracts 7200 features from each ECG signal. PCA takes 15.158 s and LDA takes 415.49 s for feature reduction from 7200 features. Table 5 shows the result of BiLSTM with other ML algorithms after the feature reduction techniques PCA and LDA. With a 98.46% accuracy rate and a 0.089 s shorter prediction time, random forest with PCA performs better in this table. Fig. 7 shows the confusion matrix of BiLSTM, Random forest with PCA and BiLSTM, Random forest

with LDA techniques. We can suggest BiLSTM Deep Learning model for feature extraction, Random Forest Machine Learning algorithm for classification and PCA for feature reduction for arrhythmia prediction compared to LSTM with Random Forest and PCA. BiLSTM with Random Forest and PCA provides an accuracy of 98.46% where as LSTM with Random Forest and PCA provides an accuracy of 98.2% only which are given in Tables 3 and 5 respectively.

This work is also validated using the K-fold cross validation approach, with K equal to ten. The results of the 10-fold cross validation technique are displayed in Table 6. The 10-fold cross validation technique produces an average accuracy of 98.30 percent for the BiLSTM, Random forest with PCA approach, which is greater than the average accuracy of the LSTM, Random forest with PCA approach which again confirms the stability of the proposed models. Even though the accuracy of the LSTM with Random forest and BiLSTM with Random forest as shown in Tables 2 and 4, declined marginally when the PCA feature reduction methodology was used, the techniques' testing time was lowered. As the result, the suggested architecture employs BiLSTM with Random Forest and PCA feature reduction techniques for the feature extraction and classification of arrhythmia problem.

5. Conclusion

This work develops a hybrid model for the automatic feature extraction and classification of various arrhythmias which includes Non-ectopic beats (normal beat), Ventricular ectopic beats, Supraventricular ectopic beats, Fusion Beats and Unknown Beats by comparing various algorithms for feature extraction, feature reduction and classification. BiLSTM provides the greatest accuracy of 96 percent when comparing the LSTM, GRU, and BiLSTM algorithms for both feature extraction and classification. The results obtained indicates that the deep learning

Table 3
Results of LSTM, PCA, LDA with Other ML algorithms.

Methods	Classes	Sensitivity	Specificity	F1-score	Accuracy	Testing time
PCA + KNN	Normal	0.91	0.96	0.93	95.86%	9.58 S
	SVE	0.95	0.95	0.95		
	VEB	0.93	0.96	0.96		
	Fusion	0.98	0.95	0.96		
	Unknown	0.99	0.94	0.99		
PCA + Naive Bayes	Normal	0.46	0.60	0.55	58%	0.015 S
	SVE	0.16	0.68	0.25		
	VEB	0.70	0.54	0.62		
	Fusion	0.78	0.52	0.78		
	Unknown	0.80	0.52	0.57		
PCA + SVM with Polynomial kernel	Normal	0.28	0.71	0.44	63.34%	12.09 S
	SVE	0.24	0.73	0.40		
	VEB	0.81	0.58	0.89		
	Fusion	0.83	0.58	0.90		
	Unknown	0.1	0.54	0.52		
PCA + SVM with Radial basis function	Normal	0.95	0.97	0.94	96.98%	7.76 S
	SVE	0.94	0.97	0.96		
	VEB	0.98	0.96	0.98		
	Fusion	0.96	0.97	0.97		
	Unknown	0.99	0.96	0.99		
PCA + SVM with Sigmoid kernel	Normal	0.71	0.79	0.67	78%	6.36 S
	SVE	0.73	0.79	0.78		
	VEB	0.74	0.78	0.71		
	Fusion	0.79	0.77	0.84		
	Unknown	0.90	0.75	0.93		
PCA + Decision tree	Normal	0.86	0.97	0.90	95.42%	0.002 S
	SVE	0.96	0.95	0.94		
	VEB	0.94	0.95	0.95		
	Fusion	0.99	0.94	0.98		
	Unknown	0.99	0.94	0.99		
PCA + Random forest	Normal	0.96	0.98	0.97	98.2%	0.090 S
	SVE	0.97	0.98	0.98		
	VEB	0.98	0.98	0.98		
	Fusion	0.99	0.98	0.99		
	Unknown	1.0	0.97	0.99		
LDA + KNN	Normal	0.90	0.97	0.92	96.32%	0.175 S
	SVE	0.96	0.96	0.96		
	VEB	0.95	0.96	0.96		
	Fusion	0.98	0.95	0.98		
	Unknown	0.99	0.95	0.99		
LDA + Naive Bayes	Normal	0.95	0.95	0.91	95.56%	0.002 S
	SVE	0.90	0.96	0.94		
	VEB	0.94	0.95	0.96		
	Fusion	0.97	0.95	0.97		
	Unknown	0.99	0.94	0.99		
LDA + SVM with Polynomial kernel	Normal	0.97	0.89	0.82	91.18%	0.190 S
	SVE	0.83	0.93	0.90		
	VEB	0.83	0.93	0.90		
	Fusion	0.92	0.90	0.95		
	Unknown	0.99	0.89	0.99		
LDA + SVM with Radial basis function	Normal	0.92	0.97	0.92	96.16%	2.79 S
	SVE	0.94	0.9	0.96		
	VEB	0.96	0.96	0.96		
	Fusion	0.97	0.95	0.98		
	Unknown	0.99	0.95	0.99		
LDA + SVM with Sigmoid kernel	Normal	0.87	0.96	0.89	94.74%	0.208 S
	SVE	0.93	0.95	0.94		
	VEB	0.9	0.94	0.96		
	Fusion	0.97	0.94	0.97		
	Unknown	0.99	0.93	0.99		
LDA + Decision tree	Normal	0.91	0.98	0.93	96.86%	0.001 S
	SVE	0.97	0.96	0.96		
	VEB	0.95	0.97	0.96		
	Fusion	0.99	0.96	0.99		
	Unknown	0.99	0.96	0.99		
LDA + Random forest	Normal	0.95	0.98	0.95	97.24%	0.052 S
	SVE	0.97	0.97	0.98		
	VEB	0.95	0.98	0.96		
	Fusion	0.99	0.96	0.99		
	Unknown	0.99	0.96	0.98		

Table 4
Results of BiLSTM with Other ML algorithms.

Methods	Classes	Sensitivity	Specificity	F1-score	Accuracy	Testing time
KNN	Normal	0.87	0.96	0.91	94.82%	373.64 S
	SVE	0.96	0.94	0.94		
	VEB	0.92	0.95	0.95		
	Fusion	0.98	0.93	0.96		
	Unknown	0.99	0.93	0.98		
Naive Bayes	Normal	0.31	0.72	0.41	64.46%	1.27 S
	SVE	0.68	0.63	0.68		
	VEB	0.60	0.65	0.69		
	Fusion	0.67	0.63	0.73		
	Unknown	0.93	0.57	0.67		
SVM with Polynomial kernel	Normal	0.88	0.95	0.89	93.9%	314.08 S
	SVE	0.95	0.93	0.92		
	VEB	0.92	0.94	0.96		
	Fusion	0.94	0.93	0.96		
	Unknown	0.97	0.92	0.98		
SVM with Radial basis function	Normal	0.91	0.94	0.89	94.04%	291.45 S
	SVE	0.92	0.94	0.94		
	VEB	0.95	0.93	0.95		
	Fusion	0.92	0.94	0.93		
	Unknown	0.98	0.92	0.98		
SVM with Sigmoid kernel	Normal	0.68	0.76	0.61	74.58%	398.01 S
	SVE	0.71	0.75	0.73		
	VEB	0.62	0.77	0.69		
	Fusion	0.78	0.73	0.79		
	Unknown	0.92	0.70	0.92		
Decision tree	Normal	0.87	0.97	0.91	95.72%	0.03 S
	SVE	0.95	0.95	0.95		
	VEB	0.95	0.95	0.96		
	Fusion	1.0	0.94	0.98		
	Unknown	0.99	0.94	0.98		
Random forest	Normal	0.98	0.98	0.99	98.84%	0.20 S
	SVE	0.98	0.99	0.98		
	VEB	0.99	0.98	0.98		
	Fusion	1.0	0.98	0.99		
	Unknown	1.0	0.98	0.99		

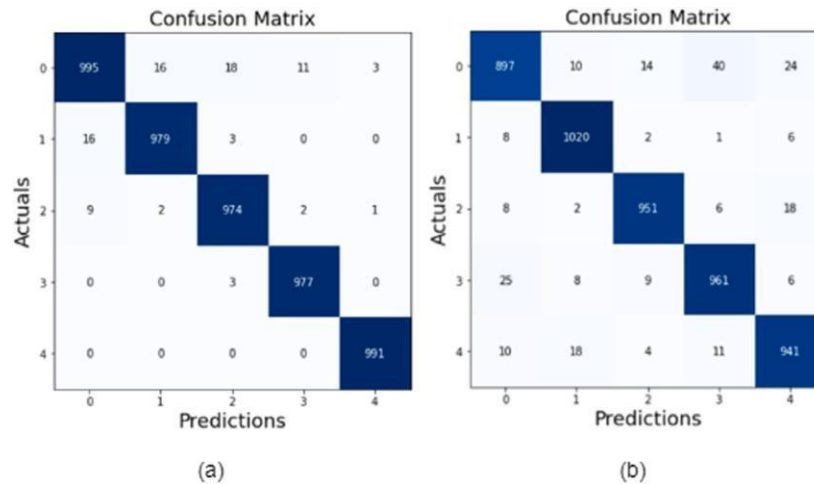


Fig. 7. Confusion matrix (a) BiLSTM with PCA + Random forest (b) BiLSTM with LDA + Random forest.

algorithm for feature extraction combined with the machine learning algorithm for classification increases the accuracy. So, for the classification of arrhythmia problem, BiLSTM combined with Random forest shows the best performance by providing best accuracy, sensitivity, specificity and f1-score. The suggested hybrid method (BiLSTM with Random forest) shows the highest accuracy of 98.84%. A better accuracy of 98.46% is produced with less prediction time by the suggested hybrid model BiLSTM + Random Forest + PCA compared to the hybrid

model LSTM + Random Forest + PCA which produces only 98.2%. Again, the same hybrid BiLSTM + Random Forest + PCA model generates a higher average accuracy of 98.30% compared to LSTM + Random Forest + PCA hybrid model with 10-fold cross validation, which again ensures the consistency of the proposed models. This proposed hybrid model is used to assist the physicians to diagnosis the heart problem patients quickly and start the treatment at the earliest possible and saves the life of patients.

Table 5
Results of BiLSTM, PCA, LDA with Other ML algorithms.

Methods	Classes	Sensitivity	Specificity	F1-score	Accuracy	Testing time
PCA + KNN	Normal	0.92	0.97	0.94	96.82%	9.87 S
	SVE	0.97	0.96	0.96		
	VEB	0.94	0.97	0.97		
	Fusion	0.99	0.96	0.97		
	Unknown	0.99	0.95	0.99		
PCA + Naive Bayes	Normal	0.54	0.72	0.60	68.66%	0.016 S
	SVE	0.64	0.69	0.75		
	VEB	0.60	0.70	0.58		
	Fusion	0.77	0.66	0.71		
	Unknown	0.86	0.64	0.78		
PCA + SVM with Polynomial kernel	Normal	0.29	0.72	0.45	64.52%	12.46 S
	SVE	0.24	0.74	0.39		
	VEB	0.81	0.60	0.90		
	Fusion	0.84	0.59	0.91		
	Unknown	0.99	0.55	0.54		
PCA + SVM with Radial basis function	Normal	0.96	0.97	0.93	97.18%	10.008 S
	SVE	0.95	0.97	0.96		
	VEB	0.99	0.97	0.97		
	Fusion	0.96	0.97	0.96		
	Unknown	0.99	0.96	0.99		
PCA + SVM with Sigmoid kernel	Normal	0.78	0.83	0.76	82.7%	5.54 S
	SVE	0.81	0.83	0.85		
	VEB	0.77	0.83	0.75		
	Fusion	0.76	0.84	0.79		
	Unknown	0.98	0.78	0.98		
PCA + Decision tree	Normal	0.86	0.97	0.90	95.14%	0.002 S
	SVE	0.96	0.94	0.95		
	VEB	0.93	0.95	0.95		
	Fusion	0.99	0.94	0.97		
	Unknown	0.99	0.93	0.99		
PCA + Random forest	Normal	0.96	0.99	0.97	98.46%	0.089 S
	SVE	0.98	0.98	0.98		
	VEB	0.98	0.98	0.98		
	Fusion	1.0	0.98	0.99		
	Unknown	1.0	0.98	1.0		
LDA + KNN	Normal	0.91	0.95	0.93	94.96%	0.21 S
	SVE	0.98	0.93	0.97		
	VEB	0.97	0.94	0.96		
	Fusion	0.94	0.95	0.94		
	Unknown	0.92	0.95	0.94		
LDA + Naive Bayes	Normal	0.94	0.94	0.90	94.46%	0.003 S
	SVE	0.98	0.93	0.98		
	VEB	0.96	0.93	0.97		
	Fusion	0.90	0.95	0.93		
	Unknown	0.92	0.95	0.94		
LDA + SVM with Polynomial kernel	Normal	0.92	0.88	0.83	89.62%	0.134 S
	SVE	0.96	0.87	0.94		
	VEB	0.92	0.88	0.94		
	Fusion	0.80	0.91	0.88		
	Unknown	0.86	0.90	0.90		
LDA + SVM with Radial basis function	Normal	0.92	0.94	0.90	94.3%	5.51 S
	SVE	0.98	0.93	0.98		
	VEB	0.96	0.93	0.97		
	Fusion	0.93	0.94	0.94		
	Unknown	0.89	0.95	0.93		
LDA + SVM with Sigmoid kernel	Normal	0.90	0.95	0.92	94.32%	0.08 S
	SVE	0.98	0.93	0.97		
	VEB	0.97	0.93	0.96		
	Fusion	0.94	0.94	0.94		
	Unknown	0.90	0.95	0.94		
LDA + Decision tree	Normal	0.91	0.95	0.92	94.52%	0.001 S
	SVE	0.98	0.93	0.97		
	VEB	0.96	0.93	0.97		
	Fusion	0.94	0.94	0.94		
	Unknown	0.91	0.95	0.93		
LDA + Random forest	Normal	0.89	0.95	0.92	94.64%	0.047 S
	SVE	0.98	0.93	0.97		
	VEB	0.98	0.93	0.96		
	Fusion	0.94	0.94	0.94		
	Unknown	0.91	0.95	0.94		

Table 6
Results of 10 fold cross validation.

Methods	Average accuracy	Standard deviation
LSTM, Random forest with PCA	98.08%	0.0021
BiLSTM, Random forest with PCA	98.30%	0.0021

CRedit authorship contribution statement

P. Varalakshmi: Conceptualization, Supervision, Project administration, Writing – review & editing. **Atshaya P. Sankaran:** Data curation, Investigation, Methodology, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study is available as open-source at <https://www.kaggle.com/taejoongyoon/mitbit-arrhythmia-database>.

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ANNEXURE II

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SREC Collaborative Innovation Center (CoIN)

CERTIFICATE OF PARTICIPATION

This is to certify that Mr. / Ms KATHIR . N of

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event titled AI ENHANCED ARRHYTHMIA DETECTION IN ECG DATA

organized by Dr./Mr./Ms. E-SHANTHINI (AP/SI-Gr) / ECE in **SREC INNOVATE 2025**

Sri Ramakrishna Engineering College, Coimbatore on 28.03.2025.

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Principal

Number - SI- 2025

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ed by Dr./Mr./Ms.ESHAANTHINI (AP/ SI.GR)/ECE. in **SREC INNOVATE 2025**

Sri Ramakrishna Engineering College, Coimbatore on 28.03.2025.


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er - SI- 2025

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project titled AI ENHANCED ARRHYTHMIA DETECTION IN ECG DATA .

mentored by Dr./Mr./Ms. E. SHANTHINI (AP / sl . Gr)/ECE in **SREC INNOVATE 2025**

held at Sri Ramakrishna Engineering College, Coimbatore on 28.03.2025.



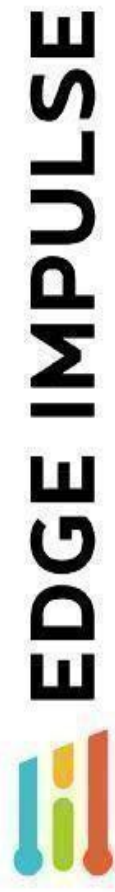
Convener - SI- 2025



SREC CoIN/- Manager



Principal



Sep 21, 2024

Kathir N

has successfully completed

Introduction to Embedded Machine Learning

an online non-credit course authorized by Edge Impulse and offered through Coursera

Shawn Hymel
Instructor
Edge Impulse

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AI & Blockchain Director, Learn2Launch
UC Berkeley

A handwritten signature in black ink, appearing to read "Kathir N".

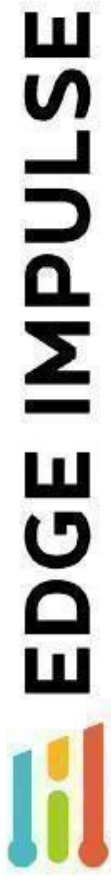
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A handwritten signature in black ink, appearing to read 'Shawn Hymel'.

Shawn Hymel
Instructor
Edge Impulse

A handwritten signature in black ink, appearing to read 'AI & Blockchain Director'.

AI & Blockchain Director, Learn2Launch
UC Berkeley

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ANNEXURE III

